The Real Effects of Investor Disagreement *

Xiaoqi Yu †

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Abstract

This paper examines how investor disagreement affects the informational feedback from stock prices to corporate investment. We show that higher investor disagreement is associated with stronger investment-to-Q sensitivity. The effect is more pronounced when managers have greater demand for external information (i.e., for R&D investment and among growth firms) and when investors' information is more effectively incorporated into prices (i.e., under stronger investor price conditioning and higher analyst coverage), providing evidence consistent with managerial learning. Additional tests indicate that the results are not explained by market timing or valuation uncertainty. Disagreement constructed from analysts' forecast revisions around earnings announcements yields similar results and traces the information flow from the market back to the firm, and analyses exploiting Regulation Fair Disclosure as an exogenous shock provide causal evidence. Overall, the findings suggest that investor disagreement can reflect heterogeneous external information that prices aggregate and managers subsequently use in investment decisions.

Keywords: Investor disagreement; revelatory price efficiency; managerial learning; corporate investment; market feedback
JEL Codes: G12, G14, G31, M41

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†Xiaoqi Yu is a PhD student at the University of Glasgow. Email: xiaoqi.yu@glasgow.ac.uk.

1 Introduction

Investor disagreement is ubiquitous in financial markets. A large literature examines its implications for asset prices and returns. When investors disagree and short-sale constraints prevent pessimistic investors from fully acting on their views, prices disproportionately reflect optimistic beliefs, causing greater disagreement to predict lower subsequent returns (Diether et al., 2002; Miller, 1977). This literature therefore focuses mainly on whether disagreement causes prices to deviate from firm fundamentals. Yet stock prices also aggregate and transmit information from financial markets to corporate decision-makers, a channel known as the market feedback effect (Goldstein, 2023). The effectiveness of this feedback depends on revelatory price efficiency, the extent to which prices reveal new information outside the firm that managers do not already possess (Bond et al., 2012). This paper examines how investor disagreement affects revelatory price efficiency and, consequently, the feedback from stock prices to corporate investment.

Disagreement and revelatory price efficiency are connected through the origin of disagreement itself. Investors disagree because they hold different private signals or interpret the same public information differently (Albagli et al., 2024; Banerjee, 2011; Kandel & Pearson, 1995). When investors trade on these heterogeneous views, prices aggregate information produced outside the firm, and greater diversity of information further strengthens this aggregation (Goldstein & Yang, 2015). Because managers already know their own information, it is precisely this externally generated component that is new to them and valuable for investment decisions. Disagreement, therefore, is not merely a friction that distorts prices; it is an observable marker of dispersed private information being incorporated into prices. To the extent that disagreement reflects such dispersed information, it should increase the information content of prices that managers can learn from. We are not aware of prior research that explores this question.

We study this question using the investment-to-Q sensitivity framework, the standard gauge of managerial learning from the market (e.g., Antoniou et al., 2023; Chen et al., 2007; Cho et al., 2025). We find that higher investor disagreement, measured by the analyst dispersion scaled by stock price, is associated with stronger investment-to-Q sensitivity. The economic magnitude is nontrivial: one interquartile increase in investor disagreement increases investment-to-Q sensitivity by 4.25%. For a one-standard-deviation increase in Q, this enhanced sensitivity translates into a 1.03% increase in investment relative to its sample mean.

We next address potential confounding by analyst coverage and measurement error in the disagreement measure. First, we control for analyst coverage and its interaction with Tobin's Q, allowing us to distinguish the effect of disagreement from the broader effects of analyst attention and information production. Consistent with Edmans et al. (2017), analyst coverage strengthens investment-to-Q sensitivity. More importantly, the effect of disagreement remains positive and significant, indicating that it captures a distinct dimension of information heterogeneity beyond analyst coverage. Second, we re-estimate the baseline model among firms followed by at least seven and at least ten analysts. These restrictions reduce measurement error and strengthen the diverse information argument as more analysts increase the scope for private information. The effect remains positive in both subsamples, with a larger point estimate among firms followed by at least ten analysts. In this subsample, an interquartile increase in disagreement raises investment-to-Q sensitivity by 7.41%. Taken together, these findings provide initial evidence that disagreement contains external information that prices reveal to managers, thereby enhancing revelatory price efficiency and facilitating managerial learning.

To strengthen identification of both the effect and the informational source of disagreement, we conduct two complementary analyses. First, we exploit Regulation Fair Disclosure (Reg FD) as an exogenous shock to the information environment underlying investor disagreement. Reg FD curtailed firms' selective disclosure of material information to favoured analysts. Before the regulation, analyst forecasts could reflect information supplied by managers and therefore already known inside the firm. After Reg FD, analysts had to rely more heavily on independent information acquisition and processing, making disagreement more likely to contain information generated outside the firm. Using matched treatment and control firms, we implement both difference-in-differences and instrumental-variable analyses. The results show that the regulation-induced increase in disagreement strengthens investment-to-Q sensitivity, providing causal evidence that disagreement enhances revelatory price efficiency.

Second, we construct a disagreement measure from analysts' forecast revisions around interim earnings announcements. Because these announcements provide analysts with a common public signal, which managers already possess, differences in subsequent revisions are more likely to reflect heterogeneous information generated outside the firm. This setting also accords with Goldstein and Yang (2019), who show that disclosure about information already known to decision-makers can help prices aggregate information on dimensions they seek to learn. We find that revision-based disagreement strengthens investment-to-Q sensitivity, supporting the view that externally generated information

embedded in disagreement enhances revelatory price efficiency. Thus, this test provides evidence that is conceptually consistent with the market feedback effect and helps establish the direction of information flow from the market to corporate investment.

To further substantiate the informational role of disagreement and the managerial learning hypothesis, we conduct several cross-sectional tests based on two conditions under which market feedback should be stronger. The first is when managers have greater demand for external information. In general, managers are well-informed about assets in place, but less informed about growth opportunities (Goldstein, 2023). Consistent with this prediction, disagreement has a stronger effect on investment-to-Q sensitivity for R&D investment and among growth firms, where investment decisions depend more heavily on uncertain future opportunities. The second is when external information is more likely to be produced and incorporated into stock prices. We find a stronger effect among firms with greater analyst coverage and a higher return variance–volume correlation. Greater analyst coverage expands the scope for independent information production (Edmans et al., 2017), while a stronger return–volume relation indicates that investors condition more heavily on prices when trading (Banerjee, 2011), allowing their heterogeneous information to be more effectively aggregated into stock prices. Together, these results support the view that disagreement strengthens managerial learning when managers need external information most and when that information is more effectively reflected in prices.

We finally consider whether the results are driven by market timing or valuation uncertainty rather than managerial learning. First, disagreement may strengthen investment-to-Q sensitivity through a market-timing channel. Under the conventional view, disagreement can generate overvaluation, which managers may exploit by issuing equity and increasing investment (Baker et al., 2003). If this mechanism drives our results, the effect should be stronger among financially constrained firms, for which overvaluation provides greater relief from external financing costs. Instead, we find that the effect is concentrated among less financially constrained firms. This pattern is inconsistent with the market-timing explanation but accords with managerial learning, as unconstrained firms are better able to adjust investment in response to new information revealed through stock prices.

Second, disagreement may strengthen investment-to-Q sensitivity because it proxies for valuation uncertainty. Greater uncertainty may induce managers to seek additional information and place more weight on all available market signals, even if disagreement does not improve revelatory price efficiency. We conduct two tests of this explanation. First, we decompose disagreement into a component explained by valuation uncertainty

and a residual component. The interaction between Q and the uncertainty-driven component is insignificant, whereas the interaction with residual disagreement remains positive and significant. Second, we examine managers' responses to peer firms' Q. If disagreement simply reflects greater uncertainty and a broader demand for information, it should also strengthen the sensitivity of investment to peer's Q (Foucault & Fresard, 2014; Yan, 2024). Although peer's Q positively predicts investment, its interaction with disagreement is insignificant. Together, these findings indicate that disagreement does not merely cause managers to rely more heavily on market signals in general. Instead, its effect is concentrated in the firm's own stock price, consistent with disagreement enhancing revelatory price efficiency.

This paper contributes to two strands of literature. First, we contribute to the literature on the market feedback effect and revelatory price efficiency. Managers learn from prices when prices reveal information generated outside the firm (Bond et al., 2012; Edmans et al., 2017), and a large body of work examines features of the information environment that shape this channel, including market structure (Foucault & Frésard, 2012), disclosure and regulation (Goldstein et al., 2023; Jayaraman & Wu, 2019), and the investor base (Antoniou et al., 2023; Cho et al., 2025; Xiong et al., 2025). These studies focus on how much external information is produced and impounded into prices. We instead identify a feature that concerns how such information is distributed across investors. When investors hold heterogeneous information about firm prospects, their disagreement marks the incorporation of external information into prices and strengthens the feedback from prices to corporate investment. Our findings show that the dispersion of information in the market, and not only its quantity, determines how useful prices are for real decisions.

Second, we contribute to the literature on investor disagreement. Existing studies primarily examine its implications for trading volume and volatility (e.g., Banerjee & Kremer, 2010; Harris & Raviv, 1993; Kandel & Pearson, 1995), as well as for valuation and expected returns (e.g., Atmaz & Basak, 2018; Diether et al., 2002; Miller, 1977; Xiouros & Zapatero, 2024). A separate line of work treats disagreement as arising from dispersed private information rather than from noise or bias (Albagli et al., 2024; Banerjee, 2011; Cookson & Niessner, 2020). We extend this informational view to corporate decisions and show that the real effect of disagreement depends on its source. Disagreement that carries heterogeneous external information enhances managerial learning, whereas disagreement associated with differences of opinion or valuation uncertainty does not. Because managers learn only from information they do not already hold, corporate investment provides a sharper test of the informational content of disagreement than pricing outcomes alone.

More specifically, our study relates to the work connecting disagreement to corporate investment. Prior studies show that disagreement affects investment through overvaluation and equity issuance (Gilchrist et al., 2005), through speculative trading and equilibrium prices (Baker et al., 2016), and through conflicting beliefs between shareholders and managers or within corporate decision-making groups (Garlappi et al., 2017; Thakor & Whited, 2011). We identify a distinct managerial learning channel in which disagreement among outside investors reflects heterogeneous external information that is incorporated into secondary-market prices and subsequently used by managers when making investment decisions.

This paper is organized as follows. Section 2 develops our main hypothesis. Section 3 describes our data and key variables. Section 4 empirically studies the impact of disagreement on revelatory price efficiency and provides causal evidence. Section 5 explores mechanism. Section 6 examines the alternative disagreement measures and the relation between disagreement and private information. Section 7 concludes.

2 Theoretical Background

A large literature shows that managers learn from stock prices when making investment decisions, a mechanism known as the market feedback effect (Bond et al., 2012; Goldstein, 2023, among others). A central insight from this literature is that what matters for real decisions is not the total amount of information in prices, but the information generated outside the firm that managers cannot extract from their own signals (Edmans et al., 2017). Managers are generally well informed about assets in place, but less so about future growth opportunities, demand conditions, and the market's assessment of firm prospects (Goldstein & Yang, 2019).

Investor disagreement provides an observable indication of how such external information is dispersed across market participants. When investors hold different private signals about firm prospects, their beliefs diverge, so disagreement is an observable counterpart of the dispersion of private information in the market (Albagli et al., 2024; Banerjee, 2011), and Cookson and Niessner (2020) document that a substantial share of investor disagreement indeed stems from differences in information sets. This information is transmitted into prices as investors trade on their signals and condition on the information prices contain (Grossman & Stiglitz, 1980; Hellwig, 1980). Crucially for managerial learning, different investors are informed about different dimensions of firm value, not merely

with different precision about the same one. Goldstein and Yang (2015) show that such diversity of information generates strategic complementarities in trading, since trading on information about one dimension reduces the uncertainty faced when trading on another. Diversity of information therefore strengthens aggregation, and because this information originates outside the firm, it is precisely the kind managers cannot obtain from their own signals.

This role of disagreement is most apparent around public disclosure. Goldstein and Yang (2019) show that disclosure about dimensions the manager already knows crowds in private information production on the dimensions the manager wishes to learn about. Disclosure thus does not exhaust the market's aggregation role; rather, investors combine the public signal with their own private assessments and reach different conclusions (Bamber et al., 1999; Kandel & Pearson, 1995; Kim & Verrecchia, 1997). In a related setting, Huang et al. (2024) show that disagreement about the quality of a public signal enhances informational price efficiency. Because managers already know what they disclose, the disagreement that emerges afterward reflects information produced outside the firm and is therefore new to them.

Investor disagreement may nonetheless arise from multiple sources and need not always reflect private information. It may arise from differences in prior beliefs (Banerjee & Kremer, 2010; Harris & Raviv, 1993), from investors believing others process information incorrectly (Fischer et al., 2022), or from valuation uncertainty and behavioral biases (Berkman et al., 2009; Daniel et al., 1998). Our argument does not require disagreement to be uniformly informational. As long as it partly captures dispersed private information produced outside the firm, it can raise revelatory price efficiency. To the extent that this component is present, disagreement strengthens managerial learning from prices.

> ***Hypothesis***: Investor disagreement is positively associated with investment-to-Q sensitivity, consistent with enhanced revelatory price efficiency.

3 Data

Our sample consists of common stocks listed on NYSE/AMEX/NASDAQ.[1] We start by obtaining accounting data from Compustat. Next, we merge this with stock prices, returns,

[1]This corresponds to stocks with share code equal to 10 or 11, and primary exchange code equal to N, A, or Q, in the CRSP database

and trading volumes from CRSP, analyst coverage data from I/B/E/S, 13F institutional holding data from LSEG S34 database. Following previous literature, we exclude financial and utility firms,[2] and exclude observations of tiny firms with stock price lower than \$5 and total assets less than \$1 million (Antoniou et al., 2023; Kacperczyk et al., 2021, among others). Since our key variable of interest is investor disagreement, measured by I/B/E/S analyst dispersion in earnings per share (EPS) forecasts, we further exclude observations with fewer than two analysts covering the stock. After the merge and filter, the final dataset consists of 5,132 unique firms for a total of 35,563 firm-year observations, covering the sample period from 1975 to 2024.

3.1 Variables

Investment Variables. Following previous literature (Antoniou et al., 2023; Chen et al., 2007; Goldstein et al., 2023, among others), we measure corporate investment (INV) using $CAPXRNDA$, $CAPXA$, or $RNDA$, depending on the specification. $CAPXRNDA$ is the sum of capital expenditure and research and development (R&D) expenses scaled by lagged total assets. $CAPXA$ and $RNDA$ are capital expenditures and R&D expenses, respectively, each scaled by lagged total assets. Tobin's Q is measured as the market value of equity plus the book value of assets minus book value of equity, scaled by total assets, and it implies firm's growth opportunity.

Investor Disagreement. This study measures investor disagreement using the dispersion in analysts' one-year-ahead EPS forecasts, calculated as the cross-sectional standard deviation of forecasts for the next fiscal year's EPS scaled by the stock price (e.g., Banerjee, 2011; Huang et al., 2021). Because our analysis is conducted at the firm-year level, we construct the annual disagreement measure by averaging the monthly dispersion measures over the firm's fiscal year. In addition, we apply a $\log(1+\cdot)$ transformation to address the high skewness of the measure.

This measure offers several advantages. Analyst dispersion is largely insulated from liquidity, trading cost, and size-related biases (Alexandridis et al., 2007), and analyst forecasts are issued frequently, providing a continuous flow of market-relevant information (Ajinkya et al., 1991). Moreover, dispersion is an appropriate proxy for disagreement when analysts reveal their unbiased expectations and investors incorporate these views

[2]Financial firms have a 1-digit SIC code of 6, while utility firms have a 2-digit SIC code of 49. Together, they account for only a small portion, approximately 2%, of our sample. Including them in the analysis yields nearly identical results.

into their beliefs (Garfinkel, 2009). In the literature, the standard deviation of analysts' EPS forecasts is often scaled by alternative denominators, such as mean absolute EPS (e.g., Diether et al., 2002), total assets (e.g., Golez and Goyenko, 2022), or book value of equity (e.g., Cen et al., 2017), to construct disagreement. A potential concern is that results may be sensitive to the choice of scaling factor. However, Cen et al. (2017) show that disagreement measures are generally robust to such choices, as the variation in the numerator primarily captures the extent of disagreement. Consistent with this view, we adopt several alternative scaling factors in our analysis for robustness, and our results remain qualitatively unchanged.

Control Variables. Our control variable includes firm size ($SIZE$), cash flow (CF), number of analyst coverage ($ANALYST$), leverage($LEVERAGE$), institutional ownership(IO), annualized cumulative returns over the next 3 years (RET), and the inverse of total assets ($INVASSET$). $SIZE$ is computed as the natural logarithm of the total assets. CF is computed as the sum of the net income before extraordinary items, and depreciation and amortization expenses, and R&D expenses, scaled by lagged total assets. $ANALYST$ is measured as the natural logarithm of the number of analyst coverage. $LEVERAGE$ is computed as the sum of debt in current liabilities and long-term debt, scaled by total assets. IO is the ratio of shares held by 13F institutional investors to total shares outstanding. RET is measured as the value-weighted market adjusted three-year cumulative return, starting from the end of the investment year. Finally, $INVASSET$ is computed as the inverse of lagged total assets, and it is included to alleviate the concern of spurious correlation since both our dependent and independent variables are scaled by lagged total assets (Chen et al., 2007).

Table 1 reports the summary statistics for our variables. All our continuous variables are winsorized at 2.5% level in each tail.[3] The means (standard deviations) of the investments, $CAPXRNDA$, $CAPXA$, and $RNDA$ are 0.137 (0.111), 0.061 (0.061), and 0.074 (0.095), respectively. This indicates that a typical firm makes annual investment equal to around 13.7% of its total assets, and this is distributed approximately evenly across capital expenditure and R&D expenses. The average Tobin's Q is 2.448. These statistics are close to typical values reported in this literature (See, e.g., Antoniou et al., 2023; Goldstein et al., 2023). Regarding disagreement, the sample firms are followed by an average of nine analysts, which generates a mean disagreement measure of about 0.004. This value corresponds to an average forecast dispersion of approximately \$0.07 per share.

[3]The winsorization at 1% level in each tail yields qualitatively similar results.

[Insert Table 1 around here]

4 The Informational Role of Investor Disagreement

4.1 Baseline Results

To examine the impact of investor disagreement ($DISA_{i,t-1}$) on revelatory price efficiency, we follow standard specification in this literature and estimate the following panel regression:

$$INV_{i,t} = \alpha_i + \gamma_t + \beta_1 Q_{i,t-1} + \beta_2 DISA_{i,t-1} + \beta_3 Q_{i,t-1} \times DISA_{i,t-1} + \theta X_{i,t-1} + \epsilon_{i,t}, \quad (1)$$

where INV_{it} firm i's investment in year t, $Q_{i,t-1}$ is firm i's Tobin's Q in year $t-1$, and $DISA_{i,t-1}$ is investor disagreement. Here, β_1 represents the average investment-to-Q sensitivity. It is expected to be positive, implying market feedback effects. The rationale is that if stock prices contain useful information not already known to managers, managers may learn from this information and use it to guide future investment decisions. Investment should therefore be sensitive to stock prices (e.g., Chen et al., 2007; Edmans et al., 2012; Goldstein, 2023). The coefficient of interest in our context is β_3, the coefficient on $Q_{i,t-1} \times DISA_{i,t-1}$. It captures the incremental impact of investor disagreement on the investment-to-Q sensitivity. Our hypothesis of the informational role of disagreement predicts a positive β_3.

$X_{i,t-1}$ is a set of control variables for firm characteristics. First, firm size correlates with its investment opportunities (Foucault & Fresard, 2014; Jayaraman & Wu, 2019), we control for size, both on its own and its interaction with Tobin's Q ($SIZE$ and $SIZE \times Q$). Second, to account for the impacts of financial constraints, we include cash flow (CF). The cash flow is well documented to affect investment (Fazzari et al., 1988), so we include it both separately and in interaction with disagreement variables (CF and $CF \times DISA$). Then, overvalued firms tend to invest more (Baker et al., 2003), so we include the annualized cumulative return over the future three years (RET). Last, since we scale both investments and Q by lagged total assets, we include the multiplicative inverse of assets ($INVASSET$) to ensure our result is not driven by the common deflator (Chen et al., 2007). Finally, we include both firm (α_i) and time (γ_t) fixed effects to account for any unobserved time-invariant firm-specific factors and variation in investment over time. The standard errors

are clustered at firm level.

We tabulate the baseline results in Table 2. Column (1), without control variables, shows that investment is positively associated with Q, with the estimated coefficient of 0.016, significant at less than 1% level. This is consistent with the managerial learning hypothesis. It suggests that managers adjust corporate investment in response to information reflected in stock prices. More importantly, the coefficient for interaction term, $Q_{i,t-1} \times DISA_{i,t-1}$, is positive as 0.569, significant at less than 1% level. This indicates that investor disagreement enhances managerial learning, consistent with our hypothesis. In Column (2), standard control variables are included, our result, though weaker, still remains. In terms of economic significance, as disagreement increases from its first (0.0005) to third (0.0042) quartile, the investment-to-Q sensitivity increases from 0.0201 to 0.0210, marking a 4.25% increase. Accordingly, a one-standard-deviation increase in Q is associated with an increase of 3.32 (3.46) percentage points in real investment among firms in the first (third) quartile of disagreement. The resulting difference of 0.14 percentage points is equivalent to 1.03% of average investment in our sample.

[Insert Table 2 around here]

We further include the natural logarithm of number of analyst coverage ($ANALYST_{i,t-1}$) and its interaction with Q ($ANALYST_{i,t-1} \times Q_{i,t-1}$) as control variable in Column (3). Since disagreement reflects dispersion in analysts' opinions, controlling for coverage helps disentangle whether the positive learning effect arises from the diversity of opinions or simply from greater analyst attention and information production. We find that analyst coverage has positive impact on investment-to-Q sensitivity, with an coefficient of 0.003 and significant at less than 1% level. This is consistent with findings in Edmans et al. (2017), who argue that greater analyst coverage enhances the RPE by improving information production and dissemination. More importantly, the coefficient for the interaction $Q_{i,t-1} \times DISA_{i,t-1}$ is higher as 0.262 and more significant after controlling for impact of analyst coverage. This indicates that the informational value embedded in investor disagreement is not subsumed by analyst coverage. Rather, disagreement captures an additional dimension of information diversity that independently strengthens managers' ability to extract useful signals from market prices.

To further substantiate our argument that disagreement carries useful outside information derived from dispersed private information among investors for managers to learn from, we repeat the baseline regression analysis in Column (3) of Table 2 by restricting the

sample based on the number of analyst coverages. This approach also helps alleviate concerns regarding measurement error in investor disagreement. Specifically, the disagreement measure can technically be computed with as few as two analyst forecasts. However, in such cases, the measure may capture estimation noise rather than genuine information diversity, as the limited number of forecasts may not provide sufficient information content. The prediction is that disagreement should have at least an equivalent, if not stronger, impact. In other words, if disagreement indeed carries information, the coefficient on the interaction term $Q_{i,t-1} \times DISA_{i,t-1}$ should be no smaller, since greater analyst coverage should improve the precision of the disagreement measure and broaden the scope of private information it captures. First, we restrict our sample to stock-year observations with at least seven analyst coverages, which is equivalent to being above the median level of coverage. The result is presented in Column (4). The coefficient for the interaction term is 0.281, significant at 5% level. Economically, as disagreement increases from its Q1 to Q3, the investment-to-Q sensitivity increases by 5.05%, from 0.0202 to 0.0212. Second, we further restrict sample to stocks covered by at least ten analysts. The coefficient is greater at 0.500 and more significant at less than 1% level. Additionally, the economical impact is more pronounced. As disagreement increases from its Q1 to Q3 the investment-to-Q sensitivity increases by 7.41%. To sum, we do not aim to claim a significantly stronger effect in the restricted samples, but the results at least suggest that disagreement reflects informative diversity rather than measurement noise.

Overall, our baseline result is consistent with the hypothesis that disagreement provides useful information for managers when making investment decisions. Specifically, investment is more sensitive to growth opportunities revealed by stock prices for firms with higher levels of disagreement.

4.2 Identification

In this section, we conduct several analyses to substantiate that endogeneity is not driving our result. There are two endogeneity concerns with our baseline analysis. First, it is possible that our specification omits variables that affect the firm's investment-to-Q sensitivity and are correlated with analyst disagreement. One important factor is the number of analyst coverages, which influences both investor disagreement and investment-to-Q sensitivity. Since this variable is already included in our baseline regression, this crucial concern is largely mitigated, though not entirely eliminated.

Second, reverse causality is present. The core of our hypothesis is that information flows from the market to the firm to guide investment, i.e., market feedback effect (Edmans et al., 2012; Goldstein, 2023). However, a simultaneous reverse channel may exist. A firm's complex investment decisions generate information that flows from the firm to the market, directly shaping the level of investor disagreement. This simultaneity in information flow creates a confounding loop that we must disentangle to establish causality. We address this concern through two complementary approaches. First, we use disagreement in analyst forecast revisions around earnings announcements to conceptually isolate disagreement arising from external information processing following a common public signal. Second, we exploit Regulation Fair Disclosure (RegFD) in difference-in-differences and instrumental-variable frameworks to provide more conventional quasi-experimental identification.

4.2.1 Event Study Around Earning Announcement Dates

Our first analysis leverages analyst forecast revisions around earnings announcement (EA) dates. On each earnings announcement date, firms release information that managers are better informed, e.g., earnings, and in response analysts revise their forecasts to update their belief upon receiving new information. This setting provides a powerful natural experiment for examining the market's informational role. When analysts update their forecasts following an earnings announcement, the resulting disagreement reflects both heterogeneous interpretations of the public news and the incorporation of their distinct private information. This diversity of beliefs generates new information that, when aggregated into stock prices, creates a valuable information flow back to the firm. Managers can learn from the novel insights reflected in prices formed under greater investor disagreement and use them to guide their investment decisions.

Focusing on revision-based disagreement around earnings announcement window offers two key advantages. First, it captures disagreement that is more likely to reflect genuine informational heterogeneity rather than noise or sentiment. Revisions around earnings announcements are directly triggered by a fundamental firm-specific news shock (e.g., current earnings and managerial guidance), whereas revisions made outside the announcement windows are more likely to reflect speculation or stale information. Second, by studying the period after managers have disclosed their information, we isolate the market's reciprocal information flow to the firm. Therefore, if revision-based disagreement enhances investment-to-Q sensitivity, it constitutes strong evidence that disagree-

ment reflects useful external information or information processing that managers do not already possess and can therefore learn from when making investment decisions.

We follow prior studies (e.g., Atiase et al., 2016; Huang et al., 2024) to construct investor disagreement based on analysts' forecast revisions. For each earnings announcement date, we collect annual earnings forecasts issued by analysts within 45 days before the announcement (pre-EA) and match them with forecasts made by the same analysts within 30 days after the announcement (post-EA). The forecast revision is defined as the difference between the post-EA and pre-EA forecasts. The revision-based investor disagreement ($DISA^{rev}$) is then measured as the cross-sectional standard deviation of these revisions, scaled by the stock price. Thus, by construction, this measure captures differences in analysts' interpretations of the same earnings news (Hong & Stein, 2007; Huang et al., 2024), reflecting heterogeneous information processing. It is well suited to our setting as it isolates disagreement that embodies new and market-generated information relevant for managerial learning.

We compute this measure for each interim earnings announcement, excluding the final fiscal-quarter announcement because forecasts issued before and after it generally pertain to different fiscal years. To align with our annual data frequency, we take the average of the three quarterly observations within each fiscal year.[4] The results remain qualitatively similar. We then replace the disagreement measure in the baseline regression with the revision-based disagreement, $DISA^{rev}_{i,t-1}$, and re-estimate the model.

We tabulate the results in Table 3. The coefficients on the interaction term, $Q_{i,t-1} \times DISA^{rev}_{i,t-1}$, remain statistically significant at the 1% level across all specifications. In Column (3), which includes the full set of control variables, the coefficient is 0.512, indicating a 9.44% increase in investment–Q sensitivity as revision-based disagreement increases from its first to third quartile. In Column (4), we further restrict the sample based on analyst coverage. Because the construction of revision-based disagreement requires each stock–quarter observation to have analysts issuing both pre- and post-EA forecasts, the number of observations and average analyst coverage are lower. To ensure sufficient coverage, we limit the sample to stocks followed by at least five analysts. The interaction coefficient is 0.648, corresponding to a 14.11% increase in investment–Q sensitivity as revision disagreement increases from its first to third quartile. Accordingly, a one-standard-deviation increase in Q is associated with an increase equivalent to 4.18% of average investment in the sample. Notably, the economic impact is much larger than that from the

[4]We also merge the most recent fiscal quarter's disagreement measure into the annual dataset and re-estimate the regression using quarterly data.

baseline, which is 1.03%. This suggests that the informational role of disagreement is more pronounced when disagreement is measured through analysts' incremental responses to a common public signal, allowing its timing and link to external information processing to be established more clearly.

[Insert Table 3 around here]

In addition, we examine whether the effect of revision-based disagreement differs depending on whether analyst forecasts converge or diverge around earnings announcements. We partition the sample according to whether forecast dispersion declines from the pre- to post-announcement period. Forecast convergence and revision-based disagreement capture distinct dimensions of belief updating. Analysts may revise their forecasts by different amounts even when their post-announcement forecasts become more similar. A decline in forecast dispersion therefore indicates that the public disclosure resolves part of analysts' prior uncertainty, while dispersion in revisions captures heterogeneity in how they incorporate the disclosure together with their private information (Harris & Raviv, 1993; Kandel & Pearson, 1995). In this setting, revision-based disagreement is more likely to reflect disciplined but heterogeneous information processing rather than unresolved uncertainty, providing a cleaner indication of external information that may enhance revelatory price efficiency (Goldstein & Yang, 2015). By contrast, an increase in forecast dispersion does not necessarily imply uninformed disagreement, but it provides a less clean setting because analysts' posterior assessments remain more dispersed after the disclosure. We therefore predict that the positive association between $DISA^{rev}_{i,t-1}$ and investment-to-Q sensitivity will be more pronounced when analyst forecasts converge following the earnings announcement.

We report the result in Columns (5) and (6) in Table 3. First, we find earnings announcement reduces analysts' disagreement about future earnings in general. In our sample, disagreement declines in roughly 74% of cases. Second, in the subsample where post-EA disagreement is lower than pre-EA, the interaction coefficient $Q_{i,t-1} \times DISA^{rev}_{i,t-1}$ is 0.504 and statistically significant at less than 1% level. By contrast, in the subsample where post-EA disagreement is higher, the coefficient is much lower as 0.182 and insignificant. This pattern is consistent with our prediction that revision-based disagreement is more strongly associated with investment-to-Q sensitivity when heterogeneous revisions occur alongside an overall reduction in posterior disagreement. In this setting, revision-based disagreement provides a cleaner indication of informative belief updating and external information incorporated into stock prices.

Overall, the earnings-announcement setting provides cleaner evidence that disagreement contains external information relevant to corporate investment. First, it focuses on analysts' responses to a clearly identified firm disclosure and provides a clear temporal sequence from managerial disclosure to market response and subsequent corporate decisions. This helps alleviate concerns that disagreement merely reflects managers' prior investment choices. Second, the stronger effect in the convergence subsample is more consistent with revision-based disagreement capturing disciplined and informative belief updating in response to the common disclosure. Together, these findings strengthen the interpretation that disagreement reflects external information incorporated into stock prices and subsequently used by managers in investment decisions.

4.2.2 DiD and IV analysis using RegFD

Our second identification strategy exploits Regulation Fair Disclosure (Reg FD), implemented by the SEC on October 23, 2000, as an exogenous shock to the analyst information environment. Prior to Reg FD, firms could selectively disclose material nonpublic information to favored analysts and institutional investors. This practice allowed management to share earnings guidance, strategic plans, and other material information through private channels, while the broader investing public had no access to the same information until it was reflected in analyst outputs or subsequent public filings. Reg FD prohibited this practice, requiring that any material disclosure made to analysts be simultaneously made available to the general public (See, e.g., Francis et al., 2006; Jorion et al., 2005).

This regulation is well suited to our setting for two related reasons. First, and more fundamentally, it altered the source of the information that analysts impound into their forecasts and, through trading, into prices. As Bond et al. (2012) and Edmans et al. (2017) emphasize, what drives the sensitivity of investment to Q is not the total information in prices but the component produced outside the firm, since information already known to managers carries no incremental value for their decisions. Before Reg FD, analysts could receive selective guidance from management, which may have anchored their forecasts around a common management-provided signal. To the extent analyst outputs were shaped by selective guidance, the information incorporated into prices was less clearly external to the firm. After the regulation, this private channel was closed, and analysts had to acquire and interpret publicly available information independently, using their own models and research. The disagreement that emerged in this environment is therefore produced outside the firm and is more likely to contain information and interpretations

not fully possessed by managers, precisely the external, revelatory information that is relevant for real investment decisions.

Second, this shift in source raised the level of analyst disagreement. Because analysts had to form forecasts from independent information rather than from common management guidance, Reg FD increased the dispersion of their forecasts. Importantly, this increase reflects, at least in part, informed differences in interpretation rather than a deterioration in the information environment. Bailey et al. (2003) document that forecast dispersion rose significantly after Reg FD while consensus forecast accuracy remained largely unchanged, which indicates that analysts continued to reach forecasts of similar quality through independent informational paths rather than producing noisier estimates. Had the higher dispersion reflected a degraded information environment, consensus accuracy should have deteriorated as well. Taken together, these two features make Reg FD a natural experiment for our hypothesis. If informed disagreement enhances revelatory price efficiency, then firms most affected by the regulation, those whose analysts shifted most from guided consensus toward independent research, should exhibit stronger investment-to-Q sensitivity in the post-regulation period.

To test this prediction, we employ a difference-in-differences (DiD) design combined with propensity score matching. The event year is set to 2000, and the estimation window spans five years on each side, covering fiscal years 1995 to 2005. Since Reg FD was implemented in October 2000, fiscal year 2000 straddles the pre- and post-regulation periods for most firms and is therefore dropped from the main DiD estimation. The post indicator ($Post$) equals one for fiscal years 2001 through 2005. Treatment ($Treat$) is defined using cross-sectional variation in pre-Reg FD forecast accuracy. The rationale follows the identification logic of Gintschel and Markov (2004), who show that the impact of Reg FD on the informativeness of analyst outputs varies systematically with pre-regulation characteristics related to the level of selective disclosure. Firms whose analysts were unusually accurate before the regulation were plausibly those receiving substantial private management guidance, and thus those most affected when that guidance channel was shut down. Mohanram and Sunder (2006) provide direct evidence for this interpretation, finding that analysts with preferential access to management enjoyed higher forecast accuracy in the pre-FD period but lost this advantage after the regulation. Forecast accuracy is measured following Huang et al. (2024). For each firm, we first compute the absolute forecast error as the absolute difference between the consensus analyst earnings forecast and actual earnings, scaled by the stock price. Then, we take its negative, so that higher values denote greater accuracy. Using the fiscal year 1999 cross-section, $Treat$ is set to one if the firm's

forecast accuracy is above the sample median, and zero otherwise.

Since treated firms may differ systematically from control firms along dimensions that independently affect investment-to-Q sensitivity, we employ propensity score matching (PSM) to ensure comparability. We estimate a logit model where the dependent variable is the treatment indicator and the covariates include firm size, Tobin's Q, analyst coverage, cash flow, leverage, institutional ownership, future stock returns, and the inverse of total assets, all measured in 1999. We then match each treated firm to a control firm using the nearest neighbor matching method. Panel A of Table 4 shows that before matching, treated and control firms differ significantly along most observable dimensions. Panel B confirms that after matching, the differences are economically small and statistically insignificant across all covariates, indicating that the matching procedure successfully creates a comparable control group.

We estimate two DiD specifications on the matched sample. The first examines whether RegFD increased investor disagreement for treated firms, which serves as a necessary condition for the identification strategy. If the regulation did not differentially affect disagreement between treated and control firms, there would be no basis for attributing any subsequent change in investment-to-Q sensitivity to the disagreement channel. Specifically, we estimate:

$$DISA_{it} = \alpha_i + \gamma_t + \beta_1 Treat_i \times Post_t + \delta X_{i,t-1} + \epsilon_{it}, \tag{2}$$

where $DISA_{it}$ is investor disagreement for firm i in year t. Our second specification examines whether the RegFD-induced increase in disagreement translated into stronger managerial learning from market prices:

$$INV_{it} = \alpha_i + \gamma_t + \theta_1 Treat_i \times Post_t \times Q_{i,t-1} + \theta_2 Q_{i,t-1} + \theta_3 Treat_i \times Post_t + \delta X_{i,t-1} + \epsilon_{it}, \tag{3}$$

where INV_{it} is investment variables, measured by $CAPXRNDA$, for firm i in year t, $Q_{i,t-1}$ is Tobin's Q in year $t-1$. $X_{i,t-1}$ is a set of control variables. In both equation, $Treat_i$ is equal to one for firms with pre-Reg FD forecast accuracy above the sample median, and zero for matched control firms. $Post_t$ is equal to one for the period following the treatment year (i.e., years 2001 through 2005) and zero otherwise. The terms $Treat_i$ and $Post_t$ are absorbed by the firm (α_i) and year (γ_t) fixed effects, respectively. The standard errors are clustered at the firm level.

In Equation 2, a positive β_1 would confirm that treated firms experienced a significant increase in disagreement following RegFD relative to control firms. In Equation 3, θ_1

captures the differential change in investment-to-Q sensitivity for treated firms relative to matched control firms in the post-regulation period. A positive θ_1 would indicate that the regulation-induced increase in informative disagreement strengthens the market feedback effect, consistent with our hypothesis. Together, β_1 and θ_1 trace out the causal chain from the regulation to disagreement and from disagreement to investment-to-Q sensitivity.

The results are presented in Panel C of Table 4. Column (1) examines the effect of Reg FD on investor disagreement. The coefficient on $Treat_i \times Post_t$ is 0.0012 and statistically significant at the 1% level, indicating that treated firms experienced a greater increase in analyst disagreement following Reg FD than matched control firms. This finding is consistent with the removal of selective managerial guidance leading analysts to rely more heavily on independent information acquisition and processing. Column (2) examines whether firms more affected by Reg FD subsequently exhibited stronger investment-to-Q sensitivity. The coefficient on $Treat_i \times Post_t \times Q_{i,t-1}$ is 0.0060 and statistically significant at the 5% level. This result indicates that the investment sensitivity to market valuations increased more strongly for treated firms after Reg FD, consistent with regulation-induced disagreement strengthening the informational feedback from stock prices to corporate investment.

[Insert Table 4 around here]

To validate the parallel trends assumption underlying our DiD design, we estimate a dynamic specification that replaces the single post indicator with a full set of year-by-treatment interactions. We use fiscal year 2000 (the event year) as the transition year and omit fiscal year 1999 as the base category. The coefficients for the pre-period year-by-treatment interactions test whether treated and control firms were already diverging before Reg FD, while the post-period coefficients trace out the treatment effect over time.

Table 5 presents the dynamic estimates. In Column (1), for disagreement, the pre-period coefficients at $t = -5$ through $t = -2$ are all statistically insignificant. This confirms that treated and control firms exhibited similar trends in disagreement before the regulation. The effect appears at $t = 0$ with a coefficient of 0.0010 and a t-statistic of 2.42, and remains positive and significant throughout the post-period, indicating a persistent increase in disagreement for treated firms. In Column (2), for the investment-to-Q sensitivity, the pre-period coefficients for Q interaction terms are all insignificant. The parallel trends assumption is therefore well supported. The treatment effect emerges at $t = +2$ with a coefficient of 0.0137 (t=3.06), and strengthens over subsequent years, reaching 0.0179 (t

= 3.90) at $t = +5$. The delayed onset of the investment-to-Q effect is economically intuitive. While the information environment changed immediately upon the regulation's implementation, the translation of this richer information set into altered managerial investment behavior plausibly requires time, as managers gradually learn to extract and act upon the more diverse signals embedded in market prices. Figure 1 plots the dynamic coefficients with 95% confidence intervals. Panel A displays the disagreement, showing a clear level shift at the event date with flat pre-trends. Panel B displays the investment-to-Q sensitivity, where the pre-event coefficients cluster tightly around zero before rising in the post-period.

[Insert Table 5 around here]

[Insert Figure 1 around here]

In our case, an important concern with the DiD analysis is that Reg FD may have simultaneously affected analyst coverage, which our baseline analysis in Table 2 shows also has a positive impact on investment-to-Q sensitivity. If treated firms experienced differential changes in coverage after Reg FD, the observed effect on investment-to-Q sensitivity might operate through the coverage channel rather than through disagreement. To address this concern, we estimate a dynamic specification with analyst coverage as the dependent variable. Figure 2 plots the year-by-treatment coefficients for analyst coverage on the matched sample. The pre-period coefficients are insignificant, and importantly, the post-period coefficients are also insignificant. This evidence indicates that Reg FD did not induce differential changes in analyst coverage between treated and control firms in our matched sample, supporting the interpretation that the observed effects on investment-to-Q sensitivity operate through the disagreement channel rather than through changes in coverage.

[Insert Figure 2 around here]

Next, we complement the reduced-form DiD analysis with an instrumental variable (IV) approach that directly estimates the causal effect of disagreement on investment-to-Q sensitivity. Following the framework of Antoniou et al. (2023), we treat both disagreement ($DISA$) and its interaction with Tobin's Q ($Q \times DISA$) as endogenous variables. Our excluded instruments are $Treat \times Post$ and $Treat \times Post \times Q$, which capture the exogenous variation in disagreement induced by Reg FD. The first-stage regressions model $DISA$ and

$Q \times DISA$ as functions of these instruments plus the full set of controls, and firm and year fixed effects,

$$DISA_{it} = \alpha_i + \gamma_t + \pi_1 Treat_i \times Post_t + \pi_2 Treat_i \times Post_t \times Q_{i,t-1} + \pi_3 Q_{i,t-1} + \delta X_{i,t-1} + \epsilon_{it}, \quad (4)$$

$$Q_{i,t-1} \times DISA_{it} = \alpha_i + \gamma_t + \phi_1 Treat_i \times Post_t + \phi_2 Treat_i \times Post_t \times Q_{i,t-1} + \phi_3 Q_{i,t-1} + \delta X_{i,t-1} + \nu_{it}. \quad (5)$$

The second stage then uses the predicted values to estimate the causal impact on investment. This IV framework parallels the approach of Antoniou et al. (2023), who instrument ETF ownership and its interaction with Q using BlackRock's acquisition of iShares as an exogenous shock, and estimate the causal effect of ETF ownership on investment-to-Q sensitivity.

Table 6 reports the IV results on the matched sample. Columns (1) and (2) present the first-stage regressions. $Treat \times Post$ significantly predicts both $DISA$ with a coefficient of 0.0013 (t=4.74) and $Q \times DISA$ with a coefficient of 0.0027 (t=4.66). The excluded-instrument F-statistics are 13.24 and 13.96, respectively. Both statistics exceed the conventional threshold of 10, indicating that the instruments are sufficiently strong. We further validate the first-stage parallel trends by plotting the dynamic coefficients for both $DISA$ and $Q \times DISA$ in Figure 3. Both panels show flat pre-period coefficients and clear post-regulation shifts, confirming that the instruments satisfy the parallel trends requirement.

Column (3) reports the second-stage estimate. The coefficient on the instrumented $Q \times DISA$ is positive at 23.0637 and statistically significant at the 10% level, while the coefficient on instrumented $DISA$ is statistically insignificant. Consistent with the reduced-form DiD evidence, this result indicates that the exogenous variation in disagreement induced by Reg FD strengthens investment-to-Q sensitivity. The IV analysis therefore provides evidence consistent with a causal effect of investor disagreement on the informational feedback from stock prices to total corporate investment.

[Insert Table 6 around here]

[Insert Figure 3 around here]

Overall, the Reg FD analysis provides causal evidence that investor disagreement enhances revelatory price efficiency. The regulation exogenously increased informative disagreement for treated firms by eliminating private management guidance, and this increase is associated with a stronger sensitivity of investment to market valuations. The

effect operates through the disagreement channel rather than through changes in analyst coverage, and it is concentrated in R&D investment, precisely where managerial learning from market prices is most valuable. These findings, combined with the revision-based evidence in Section 4.2.1, provide a consistent body of causal evidence supporting our main hypothesis.

5 Mechanism Analyses

5.1 Managerial Learning

The baseline analysis shows that investment-to-Q sensitivity increases with investor disagreement. This finding is consistent with disagreement enhancing revelatory price efficiency because prices formed under greater informational disagreement appear to contain more decision-relevant information for managers. The identification analyses strengthen this interpretation by showing that the relation persists when disagreement is more plausibly generated through external information acquisition and processing. In this section, we provide further evidence on the managerial learning channel by examining settings in which learning from prices should be more pronounced. Specifically, we consider whether the positive relation between disagreement and investment-to-Q sensitivity is stronger when managers have a greater need or ability to learn from external information and when the heterogeneous private information underlying disagreement is more effectively incorporated into stock prices.

5.1.1 Firm-Side Demand for Market Information

Managers are typically well informed about assets in place but less informed about future growth opportunities, demand conditions, and other forward-looking factors. Market prices, in contrast, aggregate dispersed private information from investors about these uncertain prospects (Goldstein, 2023; Goldstein & Yang, 2019). Thus, when disagreement, our proxy for the dispersion of private beliefs, is high, prices should convey more useful external information for managers. Guided by this logic, we conduct cross-sectional analyses exploring settings where reliance on market information is expected to vary, including 1) investment composition (CAPEX versus R&D), and 2) firm valuation type (value versus growth).

First, we distinguish between capital expenditure and R&D investment. R&D investment is largely driven by uncertain, forward-looking growth opportunities, where markets aggregate dispersed private signals about demand, technology, and competition that managers may lack. In contrast, capital expenditure mainly augments assets in place, about which managers possess relatively superior internal information. Consistent with this distinction, prior studies show that R&D involves greater uncertainty, more option-like payoffs, and distinct pricing by investors compared with tangible capital expenditure (Chan et al., 2001; Eberhart et al., 2004; Peters & Taylor, 2017, among others). Accordingly, if disagreement enhances revelatory price efficiency by increasing the external information reflected in stock prices, its incremental effect on investment-to-Q sensitivity should be more pronounced for R&D, where managers have a greater need to learn from that information.

We re-estimate the baseline regression using capital expenditure and R&D expenses, each scaled by total assets, as separate dependent variables. The results are reported in Table 7. Columns (1) and (2) use all observations for which the respective investment measure is available. Columns (3) and (4) using a common sample of firm-year observations for which both capital expenditure and R&D expenses are available. In the full sample, the coefficient on $Q_{i,t-1} \times DISA_{i,t-1}$ is positive and significant for both capital expenditure and R&D investment. However, the coefficient is nearly twice as large for R&D investment, at 0.202 with a t-statistic of 3.26, compared with 0.103 and a t-statistic of 2.50 for capital expenditure. The contrast becomes more pronounced in the restricted sample. The interaction coefficient for R&D remains economically similar and statistically significant at the 1% level, at 0.204 with a t-statistic of 3.29. By contrast, the coefficient for capital expenditure falls to 0.035 and becomes statistically insignificant.

Overall, the results show that the positive relation between disagreement and investment-to-Q sensitivity is concentrated more clearly in R&D investment. This pattern is consistent with the managerial learning interpretation. When investment concerns uncertain and forward-looking projects, managers have a greater need for external information. Consequently, the additional information reflected in stock prices under greater disagreement appears to play a more important role in R&D decisions than in capital expenditure.

[Insert Table 7 around here]

Second, we distinguish value and growth firms. Growth firms derive a larger share of their value from intangible assets and uncertain future opportunities, leading to greater

information asymmetry and reliance on external information compared with value firms, whose worth is primarily driven by assets in place (Fama & French, 1993; Farboodi et al., 2022, among others). Relatedly, Goldstein et al. (2023) argue that corporate disclosures contain more information about assets in place than about growth options. This distinction leaves greater scope for privately produced information reflected in stock prices to be useful for the investment decisions of growth firms. Accordingly, if disagreement captures heterogeneous external information that is incorporated into prices, the positive effect of disagreement on investment-to-Q sensitivity should be stronger among growth firms.

We split the sample into two groups based on the median value of book-to-market (BM) ratio in year $t-1$, and within each subsample we repeat the baseline analysis for total investment (the sum of capital expenditure and R&D expenses), capital expenditure, and R&D expenses separately. The results are reported in Table 8. Comparing Columns (1) and (4), where the dependent variable is total investment, we find that the coefficient on the interaction term $Q_{i,t-1} \times DISP_{i,t-1}$ is significantly positive among growth firms (0.353, $t = 2.71$), but insignificant and slightly negative among value firms. This contrast indicates that disagreement enhances the investment-to-Q sensitivity primarily for growth firms, consistent with the notion that dispersed private information embedded in prices provides more valuable external signals for firms whose value depends on future growth opportunities. Comparing Columns (2) and (5) and Columns (3) and (6), the results further show that the effect of disagreement is more pronounced for R&D expenses among growth firms, consistent with external information in prices being particularly valuable for uncertain and forward-looking investment decisions.

[Insert Table 8 around here]

Overall, our cross-sectional evidence supports the interpretation that disagreement reflects dispersed private information incorporated into stock prices, with its effect on investment-to-Q sensitivity more pronounced for R&D investment and among growth firms, where managers have a greater need to learn from external information.

5.1.2 Investor-Side Information Production and Aggregation

Having examined when managers have a greater need to learn from prices, we next turn to the investor side. For disagreement to enhance revelatory price efficiency, heterogeneous private information must first be produced by investors outside the firm, and must then

be incorporated into stock prices through trading. Greater production expands the scope for disagreement to reflect genuine informational heterogeneity rather than noise, while more effective aggregation determines whether that information is transmitted into prices from which managers can learn (Admati, 1985; Hellwig, 1980). Guided by this logic, we conduct two cross-sectional analyses, examining 1) analyst coverage, which captures the extent of external information production, and 2) the extent to which investors condition on prices, which captures how effectively dispersed information is aggregated.

First, we examine analyst coverage. Greater coverage implies a larger number of independent assessments of firm prospects and therefore greater scope for dispersed private information to arise. Greater coverage also reduces measurement error, since our disagreement measure can be computed from as few as two forecasts, in which case it may capture estimation noise rather than genuine information dispersion. While our baseline analysis restricts the sample by coverage to address this concern, we treat coverage here as a cross-sectional dimension along which the informational content of disagreement should vary systematically. Accordingly, if disagreement reflects externally produced information, its effect on investment-to-Q sensitivity should be stronger among firms followed by more analysts.

In each year, we split the sample into two groups based on the median number of analysts covering the stock, and within each subsample we repeat the baseline analysis for total investment (the sum of capital expenditure and R&D expenses), capital expenditure, and R&D expenses separately. The results are reported in Table 9. Comparing Columns (1) and (4), where the dependent variable is total investment, the coefficient on the interaction term $Q_{i,t-1} \times DISA_{i,t-1}$ is positive and significant in both subsamples but substantially larger among firms with high analyst coverage (0.554, t=2.98) than among firms with low coverage (0.208, t=2.02). The same pattern holds for the two components of investment. Overall, these results are consistent with disagreement conveying more external information when more analysts independently assess firm prospects, and they also mitigate the concern that our findings are driven by measurement error in disagreement among thinly covered stocks.

[Insert Table 9 around here]

Second, we examine how effectively the information underlying disagreement is aggregated into stock prices. Our argument suggests that disagreement should be particularly useful for managerial learning when investors regard others' signals as informative

and use market prices to infer those signals. In such settings, investors trade on heterogeneous assessments, observe the resulting price, and update their beliefs based on the information revealed by other market participants. This process allows stock prices to incorporate a broader set of external signals and become more informative in revelatory sense for managers making real investment decisions.

We examine this by drawing on the dynamic model developed by Banerjee (2011), which nests the rational expectations (RE) and differences-of-opinion (DO) approaches. In the RE setting, investors recognize that other investors possess payoff-relevant information and condition efficiently on prices to infer that information. Disagreement therefore arises alongside a process in which heterogeneous private signals are transmitted and aggregated through trading into prices. In the DO setting, investors place less weight on the information conveyed by prices and rely more heavily on their own signals or interpretations. Although their private information may still affect prices through their trades, dispersed signals are less effectively aggregated into an informative market price from which investors and managers can learn.

This distinction is closely related to the noisy rational-expectations literature, in which prices aggregate information that is dispersed across market participants (Admati, 1985; Hellwig, 1980). When investors condition on prices, the trading activity generated by heterogeneous beliefs not only creates volume but also facilitates the transmission of information across investors. Relatedly, Goldstein and Yang (2015) show that diversity in investors' information can improve price informativeness because information-based activity on one dimension of firm value can encourage the production and use of information on other dimensions. By contrast, differences-of-opinion models emphasize that investors may continue to disagree because they interpret signals differently or assign different precision to the same information (Banerjee & Kremer, 2010; Harris & Raviv, 1993; Kandel & Pearson, 1995). In these settings, disagreement may generate substantial trading without producing an equally effective aggregation of dispersed information.

This variation in information aggregation yields a direct prediction for market feedback effect. If disagreement strengthens investment-to-Q sensitivity because heterogeneous investor information is incorporated into stock prices and in turn facilitates managerial learning, its effect should be stronger in RE settings. In these settings, investors condition more heavily on prices and dispersed signals are more effectively aggregated. By contrast, investors in DO settings rely more heavily on their own beliefs, making disagreement less likely to produce an informative price signal from which managers can

learn. We therefore predict that the positive coefficient on $Q_{i,t-1} \times DISA_{i,t-1}$ will be larger among firms whose investors are more likely to condition on prices (RE).

Banerjee (2011) also proposes an empirical proxy to distinguish firms in which investors condition strongly on prices (RE) from firms in which investors make less use of prices (DO). Specifically, the proxy is the firm-level covariance between realized return variance and share turnover, denoted here as $Cov(V, T)$. The intuition is that greater disagreement increases trading volume under both regimes, but its relationship with perceived risk differs. Under DO, investors do not extract other investors' information from prices, so greater disagreement is associated with lower perceived risk and return volatility. Trading volume and volatility therefore tend to move in opposite directions, producing a low or negative $Cov(V, T)$. Under RE, investors use prices to infer others' information, making prices more sensitive to fundamental and supply shocks, implying higher perceived risk. Greater disagreement consequently raises both trading volume and return volatility, producing a higher positive $Cov(V, T)$. Thus, a higher value indicates that investors are more likely to condition on prices, whereas a low or negative value indicates less price-based learning.

Following Banerjee (2011), for each firm-year, we compute $Cov(V, T)$ as the covariance between squared daily return and daily turnover. Then, we split sample stocks into two subsamples based on the median value of $Cov(V, T)$ in each year. The subsample stocks with higher-than-median (lower-than-median) $Cov(V, T)$ are stocks follows RE (DO) in which investors are more (less) likely to condition on prices. For each subsample, we repeat the baseline estimation for total investment (the sum of capital expenditure and R&D expenses), capital expenditure, and R&D expenses separately. The results are tabulated in Table 10.

[Insert Table 10 around here]

First, in the DO subsample, the coefficient on $Q \times DISA$ is insignificant across all three investment measures (Columns 1 to 3). In the RE subsample, by contrast, the coefficient is significantly positive for total investment (Column 4). This pattern supports our argument that when investors condition more heavily on prices, the private information underlying their disagreement is more effectively aggregated into stock prices, thereby increasing revelatory efficiency. The stronger investment response to these prices is consistent with managers learning from the additional information they contain. Second, separating capital expenditure from R&D expenses within the RE subsample sharpens

this result further. Comparing Columns (5) and (6), the coefficient on $Q \times DISA$ is small and insignificant when the investment variable is *CAPXA*, but large and significant at the 1% level when the investment variable is *RNDA*. Economically, the investment-to-Q sensitivity rises by approximately 11.51% for R&D expenses as disagreement increases from its first to third quartile, compared with only 1.22% for capital expenditures. Section 5.1.1 already shows that managers are more likely to learn from market prices when making R&D investment decisions because they have less of an information advantage over these uncertain and forward-looking projects.

Taken together, the results show that the positive relation between disagreement and investment-to-Q sensitivity is strongest when managers have a greater need for external information and investors' heterogeneous private information is more effectively incorporated into prices.

5.2 Market Timing

The incremental effect of investor disagreement may also arise through a market-timing channel. A conventional implication of disagreement is that it can lead to overvaluation. This occurs when optimistic and pessimistic investors disagree, but pessimistic views are not fully reflected in prices because of short-sale constraints (Miller, 1977). Consistent with this view, many studies document a negative association between disagreement and subsequent stock returns (e.g., Bali et al., 2026; Diether et al., 2002; Hong & Sraer, 2016). If disagreement leads to overvaluation, firm managers may exploit the mispricing by issuing overpriced equity and investing the proceeds (Baker et al., 2003). Under this channel, investment becomes more sensitive to Q not because stock prices are more efficient in the revelatory sense and provide managers with information from which to learn, but because overvaluation lowers the cost of external equity, with this effect becoming stronger as disagreement increases. Thus, a positive coefficient for $Q \times DISA$ is also consistent with this market timing channel.

To assess the feasibility of this channel, we examine whether the effect of disagreement is more pronounced among financially constrained firms. Under the market-timing explanation, disagreement-induced overvaluation allows firms to issue equity on favourable terms and finance investment at a lower effective cost (Baker et al., 2003). This benefit should be particularly important for constrained firms whose investment opportunities cannot be fully financed with internal funds and whose investment therefore depends

more heavily on access to external capital (Fazzari et al., 1988). For less constrained firms, easier access to finance leaves less scope for overvalued equity to affect real investment. The market-timing explanation therefore predicts that the positive relation between disagreement and investment-to-Q sensitivity should be stronger among more financially constrained firms.

By contrast, the managerial learning channel predicts the opposite pattern. Learning from prices requires not only that stock prices contain useful information, but also that managers have sufficient financial flexibility to respond to that information. Financially constrained firms may recognize informative price signals but remain unable to adjust investment accordingly (Edmans et al., 2017). The effect of disagreement on investment-to-Q sensitivity should therefore be stronger among less financially constrained firms. Comparing the coefficient on $Q \times DISA$ across these subsamples therefore helps distinguish between the two mechanisms.

Following Edmans et al. (2017), we measure financial constraints (FC) as the difference between capital expenditures and cash flows scaled by capital expenditures, where higher (lower) values indicate greater external (internal) financing and thus lower (higher) financial constraints. We sort the stocks based on the median value of $FC_{i,t-1}$ into two subsamples in each year, and then repeat the baseline analysis and report the results in Table 11. It is observed the significantly positive coefficients for $Q \times DISA$ only show up in less financially constrained subsample, suggesting managerial learning channel, instead of market timing channel, is at play in our case.

[Insert Table 11 around here]

Beyond this empirical evidence, the two channels also differ conceptually in where they operate. As Goldstein (2023) notes, the market timing channel operates through primary markets, where firms raise new capital, whereas secondary market activity does not directly provide capital to the firm. Moreover, even in primary markets, overvaluation more readily explains equity issuance than real investment. A manager who recognizes that the firm's shares are overvalued may issue equity and retain the proceeds, since the mispricing itself does not improve the underlying profitability of the firm's investment opportunities. Our disagreement measure is constructed from analyst forecasts, which form part of the secondary market information environment, while our outcome is real investment rather than equity issuance. Both features therefore lie outside the most natural setting for the market-timing channel. By contrast, the managerial learning channel oper-

ates through the information reflected in secondary market prices and directly concerns firms' real investment decisions (Edmans et al., 2017). The market timing explanation is therefore unlikely to account for our findings on either empirical or conceptual grounds.

5.3 Valuation Uncertainty

Our finding that disagreement increases investment-to-Q sensitivity is consistent with disagreement enhancing revelatory price efficiency by incorporating external private information into prices. However, disagreement may instead proxy for underlying firm-level uncertainty. Such uncertainty can both generate greater investor disagreement and lead managers to place greater weight on stock prices when making investment decisions. This may produce a pattern resembling stronger learning from prices even if disagreement does not add information to them (Berkman et al., 2009). We therefore test whether the effect reflects the informational content of disagreement or the uncertainty associated with it.

We conduct two tests. First, we decompose disagreement into two components: a valuation-uncertainty-driven component ($\widehat{DISA}_{i,t-1}$) and an orthogonal component ($DISA^{\perp}_{i,t-1}$). In doing so, we argue that disagreement carries useful private information, while also acknowledging that it partially reflects valuation uncertainty. On the one hand, the coefficient for the interaction term ($Q_{i,t-1} \times DISA^{\perp}_{i,t-1}$) is expected to be significantly positive, consistent with our baseline result. On the other hand, if managers place greater weight on stock prices due to heightened uncertainty, we should also expect the coefficient for the interaction ($Q_{i,t-1} \times \widehat{DISA}_{i,t-1}$) to be significantly positive.

Second, we examine how investor disagreement affects managerial learning from peer firms' stock prices. Prior studies document that managers actively extract information from peers' stock prices, particularly when those prices are more informative than their own (Antoniou et al., 2023; Foucault & Fresard, 2014). In our setting, the uncertainty and informational interpretations of disagreement generate distinct predictions. If disagreement merely captures valuation uncertainty, managers facing higher uncertainty should rely more heavily on any external information source, including peer firms' prices. In this case, we would expect the interaction between disagreement and peers' Q ($PeerQ_{i,t-1} \times DISA_{i,t-1}$) to be significantly positive, reflecting broader reliance on market signals. Conversely, if disagreement represents the market's heterogeneous interpretation of firm-specific information rather than general uncertainty, its effect should be concentrated on learning from the firm's own stock price, with little or no impact on learning from peers'

prices.

5.3.1 VU-driven disagreement versus Informational Disagreement

For this first test, we consider two valuation uncertainty proxies, historical earning volatility and firm complexity. First, historical earnings provide a key basis for forecasting future performance. When a firm's past earnings are more volatile, its future earnings become harder to predict, leading investors to disagree more about the firm's fundamental value (Berkman et al., 2009). We measure earnings volatility ($EarnVol$) as the standard deviation of the ratio of the earning (Income before extraordinary items available for common) to total assets over the past five fiscal years. We require a minimum of 3-year in calculation. Second, we use Loughran and McDonald (2024)'s firm complexity as a proxy for valuation uncertainty.[5] They show that firms with more complex 10-K filings exhibit higher stock return volatility and analyst forecast dispersion, indicating greater difficulty in valuation. Such complexity naturally leads investors to interpret the same disclosures differently, generating disagreement that reflects valuation uncertainty rather than heterogeneous private information.

Next, we estimate a panel regression of investor disagreement ($DISA_{i,t-1}$) on each valuation uncertainty proxy separately, controlling for firm and year fixed effects. From this regression, we obtain the predicted component of disagreement ($\widehat{DISA}_{i,t-1}$) and the residual component ($DISA^{\perp}_{i,t-1}$). We then re-estimate the baseline model by including these two components both individually and jointly. The results are reported in Table 12, where Columns (1)–(3) correspond to earnings volatility and Columns (4)–(6) correspond to firm complexity.

The central question is whether the baseline effect is attributable to disagreement generated by valuation uncertainty. When earnings volatility is used as the valuation uncertainty proxy, Column (1) shows that the coefficient on $Q_{i,t-1} \times \widehat{DISA}_{i,t-1}$ is 0.242 and statistically insignificant. In Column (2), the coefficient on $Q_{i,t-1} \times DISA^{\perp}_{i,t-1}$ is positive at 0.575 and statistically significant at the 1% level. These results indicate that the component of disagreement not explained by earnings volatility significantly strengthens investment-to-Q sensitivity.

When the two components are included jointly in Column (3), the coefficient on $Q_{i,t-1} \times$

[5] Loughran and McDonald (2024) make their dataset publicly available at https://sraf.nd.edu/complexity/.

$DISA^{\perp}_{i,t-1}$ remains positive and significant at 0.582. However, the coefficient on $Q_{i,t-1} \times \widehat{DISA}_{i,t-1}$ also becomes positive and statistically significant at 0.461. The earnings-volatility results therefore do not entirely rule out the possibility that valuation uncertainty contributes to the stronger investment response to stock prices. Nevertheless, the persistent significance of the residual component indicates that the baseline relation cannot be explained solely by disagreement associated with earnings volatility and remains consistent with disagreement containing information beyond that captured by this uncertainty proxy.

[Insert Table 12 around here]

Columns (4) through (6) use firm complexity as an alternative proxy for valuation uncertainty. The coefficient on $Q_{i,t-1} \times \widehat{DISA}_{i,t-1}$ is statistically insignificant both when examined separately in Column (4) and when included jointly with the residual component in Column (6). By contrast, the coefficient on $Q_{i,t-1} \times DISA^{\perp}_{i,t-1}$ is positive and statistically significant in both Columns (5) and (6). The evidence based on firm complexity therefore provides stronger support for the interpretation that the positive relation between disagreement and investment-to-Q sensitivity is concentrated in disagreement not explained by valuation uncertainty. Taken together, the two proxies suggest that valuation uncertainty may contribute to the result in some specifications, but it does not fully account for the positive incremental effect of disagreement on investment-to-Q sensitivity.

5.3.2 Disagreement and Investment-to-Peer's Q sensitivity

Our second test investigates how investor disagreement affects managerial learning from peers' prices, where peers are defined using both traditional and text-based industry classifications. Specifically, we employ the Standard Industrial Classification (SIC) 2-digit system and the Fixed Industry Classification developed by Hoberg and Phillips (2016). The FIC-100 clusters firms into 100 text-based industries based on the cosine similarity of their 10-K product descriptions, providing a more economically grounded measure of product-market relatedness than SIC.[6] Unlike the administratively assigned and infrequently updated SIC codes, FIC-100 captures evolving product similarities while maintaining fixed industry definitions derived from 1997 clusters. Our analysis based on firm-level FIC-100 ranges from 1996 to 2015 due to data availability. When aggregating peers'

[6]Our results remain robust when using alternative FIC groupings such as FIC-25, FIC-50, and FIC-200.

Q, we consider three alternatives: the simple average, the value-weighted average based on firm size, and the median.

The results are presented in Table 13. Across all specifications, the coefficient on peers' Q ($PeerQ_{i,t-1}$) is significantly positive, indicating that firms tend to increase investment when their peers exhibit higher market valuations. This finding is consistent with the notion that managers learn from peers' prices and interpret them as informative signals about industry-wide prospects. However, the interaction term $PeerQ_{i,t-1} \times DISA_{i,t-1}$ provides little evidence that investor disagreement enhances managerial learning from peers' valuations. In the FIC-100 industry classifications, the interaction terms are statistically insignificant across all three measures of peers' Q. Under the SIC 2-digit industry classifications, the interaction terms are negative and statistically significant, suggesting that higher disagreement, if anything, weakens rather than strengthens firms' investment response to peers' Q.

[Insert Table 13 around here]

Overall, these findings provide more support to our interpretation that disagreement reflects informational heterogeneity embedded in a firm's own stock price rather than general valuation uncertainty. If disagreement primarily captured valuation uncertainty, managers facing greater uncertainty should rely more heavily on external valuation signals, including peers' prices, implying a positive coefficient on $PeerQ_{i,t-1} \times DISA_{i,t-1}$. Instead, we find no positive interaction effect, where the estimates are either insignificant or significantly negative. This pattern suggests that disagreement enhances learning from a firm's own price rather than from peer firms' prices. The results are robust across alternative measures of peers' Q, namely simple average, value-weighted average, and median, and across both FIC-100 and SIC-based industry classifications.

6 Additional Analyses

6.1 Alternative Disagreement Measures

Our baseline analysis measures disagreement as the standard deviation of analysts' EPS forecasts scaled by stock price. Although widely used, this measure is not without concerns. It may also capture earnings uncertainty, forecast bias, or effects of the scaling

choice. We therefore employ several alternative measures of disagreement to examine its effect on investment-to-(Q) sensitivity. The consistent results across these measures confirm the robustness of our baseline findings.

6.1.1 13F Institutional Trading-Based Measures

First, we use two institutional trading based measures, 13F buyer versus seller disagreement ($DISA^{BSD}$) and institutional trading dispersion ($DISA^{TD}$), which capture disagreement revealed through the actual trades of institutions rather than stated beliefs. These measures align closely with our mechanism. For disagreement to enhance revelatory price efficiency, differing beliefs must be traded on so that private information is impounded into prices, and disagreement expressed in institutional trading captures precisely this stage of information incorporation. We measure these two as follows:

$$DISA^{BSD}_{i,t} = -\ln\left(\left|\frac{Buyers_{i,t} - Sellers_{i,t}}{Buyers_{i,t} + Sellers_{i,t}}\right| + 0.0001\right), \tag{6}$$

$$DISA^{TD}_{i,t} = \ln\left(\frac{\mathrm{SD}_j\left(\Delta Holdings_{j,i,t}\right)}{Shrout_{i,t}}\right), \tag{7}$$

where $Buyers_{i,t}$ and $Sellers_{i,t}$ are the number of buyers and sellers on stock i in quarter t, respectively. $\Delta Holdings$ is the change of holdings, in terms of number of shares, of institution i on stock i in quarter t, and $Shrout_{i,t}$ is the split-adjusted shares outstanding of stock i in quarter t. Here, an institution is a *Buyer* (*Seller*) if its $\Delta Holdings_{j,i,t}$ is positive (negative).

The construction of $DISA^{BSD}$ is motivated by the broader literature showing that heterogeneous beliefs are revealed through differences in investors' trading decisions (e.g., Garfinkel, 2009; Goetzmann & Massa, 2005). By focusing on the relative numbers of institutional buyers and sellers, the measure captures the extent to which institutions take opposing directional positions in the same stock. The conventional absolute buy–sell imbalance, $\left|\frac{Buyers_{i,t}-Sellers_{i,t}}{Buyers_{i,t}+Sellers_{i,t}}\right|$, equals one when institutional trading is entirely one-sided, indicating directional consensus rather than disagreement. Conversely, it equals zero when buyers and sellers are perfectly balanced, indicating the greatest degree of disagreement. To ensure that higher values correspond to greater disagreement, consistent with our other measures, we take the negative natural logarithm of the absolute buyer–seller imbalance. This transformation reverses the ordering of the original imbalance measure, such that a

more balanced distribution of buyers and sellers produces a higher disagreement value, while also reducing skewness. We add 0.0001 before taking the logarithm to accommodate observations for which the absolute imbalance is zero.

The second measure is inspired by Jiang and Sun (2014) and Alldredge and Caglayan (2025). Jiang and Sun (2014) show that dispersion in active mutual fund holdings positively predicts future returns and attribute this to a differentially informed managers channel, in which informed managers place larger and more divergent bets after receiving private signals unobserved by their peers. This argument directly supports our premise, since it shows that dispersion arising from institutional trading reflects genuine private information rather than noise. Alldredge and Caglayan (2025) construct institutional trade dispersion as the cross sectional standard deviation of the percentage change in each institution's holdings, where the percentage change is computed relative to that institution's own prior holdings. Our $DISA^{TD}$ instead takes the cross sectional standard deviation of the raw change in shares held across institutions, and then scales this dispersion by total shares outstanding of the firm.[7] This choice matches our setting, since revelatory price efficiency operates through how much a given trade moves the stock price, and price impact depends on trade size relative to the total tradable float of the firm rather than on trade size relative to the trading institution's own book. Scaling by shares outstanding therefore keeps our measure aligned with the channel through which disagreement is impounded into prices and ultimately shapes investment to Q sensitivity.

The results of estimation using institutional trading based disagreement measures are reported in Column (1) and (2) of Table 14. The coefficient for the interaction term between $DISA^{BSD}$ ($DISA^{TD}$) and Q is 0.001 (0.005), and both are statistically significant at less than 1% level. Economically, as disagreement increases from its first to third quartile, the investment-to-Q sensitivity increases by approximately 6% and 59% for $DISA^{BSD}$ and $DISA^{TD}$, respectively. Recall that the investment-to-Q sensitivy increases by approximately 5% as our baseline disagreement, the standard deviation of analysts' EPS forecasts scaled by stock price, increases from its first to third quartile. Thus, the economic significance of $DISA^{BSD}$ is comparable to our baseline results, while the impact of $DISA^{TD}$ is surprisingly high. Overall, these results confirm that our baseline finding is not an artifact of relying on analyst based disagreement, since a measure built entirely from institutional trades produces the same positive relation between disagreement and investment to Q sensitivity. The larger magnitude for $DISA^{TD}$ is consistent with it capturing dispersion in trade size relative to firm float, a channel more directly tied to price impact than the buyer

[7] Alldredge and Caglayan (2025)'s construction yields qualitatively similar results.

versus seller balance underlying $DISA^{BSD}$.

[Insert Table 14 around here]

6.1.2 Alternative Scaling Factors

Second, we rescale the standard deviation of analysts' EPS forecasts by the mean forecast and by assets per share, addressing the concern that scaling by price mechanically links the measure to Q. We re-estimate Equation 1 by replacing the disagreement measure, and reporting results in Column (3) and (4) in Table 14. Both coefficients are significant at the 5% level, indicating that our baseline results are robust to the choice of scaling factor. This robustness to alternative scaling choices is also broadly consistent with the existing literature (e.g., Banerjee, 2011; Cen et al., 2017).

6.1.3 Trading Volume

Finally, we include standardized unexplained volume ($DISA^{SUV}$), a volume-based proxy for investor disagreement. Following Garfinkel (2009) and Huang et al. (2021), we first compute monthly stock turnover and remove its predictable dynamics by estimating a rolling AR(4) model of the natural logarithm of turnover. We then regress the unexplained turnover component on the positive and negative components of stock returns using a 60-month rolling window, requiring at least 36 valid monthly observations:

$$Volume_{i,m} = \alpha_i + \beta_{1,i} R^{+}_{i,m} + \beta_{2,i} R^{-}_{i,m} + \varepsilon_{i,m}, \tag{8}$$

where $R^{+}_{i,m} = \max(R_{i,m}, 0)$ and $R^{-}_{i,m} = \min(R_{i,m}, 0)$. The standardized residual from this rolling regression constitutes our monthly disagreement measure:

$$DISA^{SUV}_{i,m} = \frac{\varepsilon_{i,m}}{\sigma_{\varepsilon,i,m}}. \tag{9}$$

Higher values indicate abnormally high trading volume that cannot be explained by normal turnover dynamics or return-related volume effects and therefore capture unusually intensive trading associated with differences in investors' opinions. To align the measure with our annual sample, we define annual $DISA^{SUV}$ as the average monthly value over the 12 months up to and including the firm's fiscal year-end month, requiring at least six valid monthly observations.

The result is presented in Column (5) of Table 14. In contrast, the coefficient on the interaction between Q and $DISA^{SUV}$ is economically small and statistically insignificant. This suggests that standardized unexplained volume does not significantly strengthen the investment-to-Q sensitivity in our setting. One possible interpretation is that abnormal trading volume captures a broad mixture of trading motives, including liquidity demand, portfolio rebalancing, and speculative activity, and therefore contains a weaker informational signal than disagreement measures constructed from analysts' forecasts or institutional trading decisions.

Overall, the results indicate that our main finding is robust across several alternative disagreement measures, but not when disagreement is proxied by standardized unexplained volume.

6.2 Investor Disagreement and Private Information

We argue the positive incremental impact of disagreement on revelatory price efficiency arises because disagreement contain useful private information for managers to learn, and we have attempted to prove this using a battery of tests. Here, to provide further support, we examine whether investor disagreement is directly related to measures of private information. In our context, a positive association would suggest that disagreement contains an valuable informational component.

Chen et al. (2007) argues that managers learn from the private information in stock price by showing that investment-to-Q sensitivity is positively related to two measures of the amount of private information in stock price, namely price nonsynchronicity ($Nonsync$) and probability of informed trading (PIN). Besides these two, we also consider two additional proxy variables of private information, generalized PIN ($GPIN$) and Brogaard et al. (2022)'s private information share ($PrivateInfo$). Four measures complement each other and provide a broader assessment of different aspects of private information embedded in stock prices, thereby strengthening the robustness of our analysis.

$Nonsync$ was first proposed by Roll (1988) and has been widely used in the literature (e.g., Brogaard et al., 2022). It is computed as $\ln\left(\frac{1-R^2}{R^2}\right)$, where R^2 is obtained from a stock-level time-series regression of monthly excess returns on the Fama and French (1993) three factors. Intuitively, $Nonsync$ captures the extent to which a stock's return variation is not explained by common risk factors. A higher value indicates lower comovement with market-wide and factor-related returns, and is therefore interpreted as reflecting more

firm-specific, potentially private, information incorporated into stock prices. We estimate *Nonsync* for each stock-year using the past 60 months of returns, requiring a minimum of 36 monthly observations.

PIN was first proposed by Easley et al. (1996) as a structural market microstructure measure of information asymmetry. It uses order-flow data, especially buyer- and seller-initiated trades, to infer the likelihood that trades are driven by privately informed investors rather than uninformed liquidity traders. Intuitively, a higher *PIN* suggests that a larger fraction of trading is associated with private information, and therefore that the stock price contains more information not yet publicly available. The *GPIN* extends the original *PIN* model by allowing greater flexibility in noise trading intensity, rather than attributing high turnover too mechanically to informed trading. This distinction is important because Duarte et al. (2020) show that the original *PIN* model may partly confound private information with variation in uninformed trading volume, whereas *GPIN* is designed to better separate private information arrival from noise-driven trading activity.[8]

PrivateInfo was proposed by Brogaard et al. (2022) and measures the share of stock-return variance attributable to private firm-specific information revealed through trading. Their variance decomposition separates stock price movements into market-wide information, public firm-specific information, private firm-specific information, and noise. In this framework, private information is identified from the permanent price impact of shocks to signed trading volume, capturing the extent to which trading impounds privately held information into prices. We follow their method and estimate *PrivateInfo* separately for each stock-year using daily observations within the corresponding year. A higher *PrivateInfo* therefore indicates that a larger proportion of a stock's return variation reflects private information incorporated through trading.

We perform a portfolio-sorting analysis to examine the association between investor disagreement and private information measures. Specifically, in each year, we sort stocks into quintile portfolios based on $DISA_{i,t-1}$ and calculate the cross-sectional mean of each private information measure in year t within each portfolio. We then report the time-series average of these annual portfolio means. The high-minus-low (HML) portfolio is defined as the annual difference between the highest- and lowest-disagreement portfolios, and we report its time-series average. Statistical significance is assessed using Newey and West (1987) standard errors based on the time series of annual portfolio means. A positive

[8]We thank Stephen Brown for providing *PIN* measure (available at https://terpconnect.umd.edu/šstephenb/), and Edwin Hu for providing *GPIN* measure (available at https://edwinhu.github.io/pin/).

and statistically significant HML estimate would indicate that stocks with greater prior disagreement are associated with higher subsequent levels of private information.

We tabulate the results in Table 15. The results show a clear positive association between disagreement and private information measures. Stocks in the highest-disagreement portfolio have higher values of *Nonsync*, *PIN*, *GPIN*, and *PrivateInfo*. The HML differences are positive for all four measures and statistically significant, with particularly strong evidence for *PIN* and *GPIN*. Economically, these differences are meaningful. The HML difference in *Nonsync* of 0.241 implies that the factor model R^2 of high disagreement stocks is about 4.88 percentage points lower than that of low disagreement stocks (26.3% versus 31.1%),[9] indicating that a correspondingly larger share of their return variation is firm specific rather than driven by common factors. The spreads in *PIN* and *GPIN* indicate that the expected share of trades originating from privately informed investors is 2.5 and 2.1 percentage points higher, respectively, for high disagreement stocks, and the fact that the effect survives under *GPIN* suggests it is not an artifact of the mechanical link between turnover and PIN documented by Duarte et al. (2020). Finally, the share of return variance attributable to private information revealed through trading, as captured by *PrivateInfo*, is 1.1 percentage points higher for high disagreement stocks, although this difference is significant only at the 10% level. Taken together, the evidence from all four measures consistently indicates that stocks subject to greater investor disagreement subsequently embed more private information into their prices.

[Insert Table 15 around here]

7 Conclusion

This paper provides evidence that investor disagreement can reflect dispersed external information that stock prices aggregate and managers subsequently use in investment decisions. Investor disagreement strengthens investment to Q sensitivity, and the effect concentrates precisely where managerial learning should operate, in R&D investment, among growth firms, and where investors' heterogeneous information is more effectively incorporated into prices. Quasi experimental evidence from Regulation Fair Disclosure establishes causality, and disagreement constructed from forecast revisions around earnings

[9] They are transformed from 0.811 and 1.052, as reported in Table 15. Note that *Nonsync* is computed as $\ln\left(\frac{1-R^2}{R^2}\right)$. Therefore, the implied R^2 can be recovered as $R^2 = \frac{1}{1+e^{Nonsync}}$.

announcements, which isolates information generated outside the firm after managers have disclosed their own, traces the direction of the feedback from the market to the firm. The results are not explained by market timing or valuation uncertainty, indicating that the real effects of disagreement depend on its informational source rather than on the frictions emphasized in the conventional view.

Our findings highlight the real effects of disagreement in financial markets. Rather than representing only valuation distortions under conventional view, disagreement can reflect dispersed external information that is incorporated into prices and subsequently used by managers. This implies that managers may benefit from considering the disagreement underlying stock-price signals, particularly disagreement in analysts' forecast revisions. Our Regulation FD evidence also suggests that disclosure rules can influence real decisions by affecting how external information is produced and incorporated into prices. Policies that support timely disclosure and broad analyst coverage may therefore strengthen the informational feedback from financial markets to firms.

Our study is subject to several limitations, which in turn suggest avenues for future research. First, although the evidence indicates that the informational component of disagreement enhances revelatory price efficiency, this component cannot be observed directly. Our identification designs isolate settings in which disagreement is more plausibly informational, but they do not deliver a direct measure of it. Future research could therefore distinguish disagreement arising from heterogeneous private information from that driven by differential interpretation or behavioral noise and develop empirical proxies for each (See, e.g., Goulding et al., 2026a, 2026b). Second, our main measure captures disagreement among analysts rather than among investors, and the mapping between the two rests on the assumption that investors incorporate analyst views into their beliefs. The consistent results from measures constructed from institutional trading are reassuring, but disagreement among the broader investor base remains unobserved. Finally, investment-to-Q sensitivity is an indirect gauge of managerial learning, as it infers learning from the responsiveness of investment to prices rather than from managers' information sets. Beyond these measurement issues, future work could examine how corporate governance, managerial characteristics, and disclosure practices affect managers' ability and willingness to use information embedded in prices, which would help clarify when disagreement is most likely to strengthen market feedback and influence real decisions.

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Tables

Table 1: Descriptive Statistics

This table reports summary statistics for the main variables used in the analysis. The definitions of the variables are presented in Table A1. The main sample period is from 1975 to 2024. Due to data availability, the sample period for revision-based disagreement calculated from unadjusted detailed I/B/E/S analyst forecasts is from 1982 to 2024, and the sample period for the firm complexity measure of Loughran and McDonald (2024) is from 1995 to 2021. All variables are winsorized at the 2.5% level in each tail.

Variable	N	Mean	SD	P25	Median	P75
$CAPXRNDA$	35,563	0.137	0.111	0.058	0.104	0.179
$CAPXA$	35,563	0.061	0.061	0.022	0.043	0.079
$RNDA$	35,563	0.074	0.095	0.007	0.035	0.107
Q	35,563	2.448	1.826	1.312	1.849	2.901
$DISA$	35,563	0.004	0.008	0.001	0.001	0.003
$ANALYST$	35,563	2.004	0.751	1.386	1.946	2.639
$SIZE$	35,563	6.657	1.769	5.309	6.542	7.864
CF	35,563	0.070	0.145	0.041	0.097	0.148
RET	35,563	0.042	0.236	-0.094	0.056	0.189
$INVASSET$	35,563	0.004	0.007	0.000	0.001	0.005
BM	35,563	0.462	0.342	0.221	0.379	0.611
FC	35,539	-0.218	1.767	-0.775	-0.031	0.567
$EarnVol$	26,101	0.045	0.047	0.015	0.028	0.056
$Complexity$	22,547	0.430	0.160	0.310	0.404	0.528
$DISA^{rev}$	19,625	0.005	0.008	0.001	0.002	0.005
$ANALYST^{rev}$	19,625	1.267	0.531	0.693	1.099	1.609

Table 2: Baseline Regression: Investment-Q Sensitivity and Investor Disagreement

This table reports estimates from panel regressions of capital investment on Tobin's Q, investor disagreement, and their interaction. The investment variable in the baseline specification is the sum of capital expenditure and R&D expenses scaled by lagged total assets ($CAPXRNDA_{i,t}$). Columns (1) to (3) use the full sample, while Columns (4) to (5) restrict the sample to firms followed by more analysts. T-statistics, based on standard errors clustered at the firm level, are reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

	Full Sample			No. of Analyst ≥ 7	No. of Analyst ≥ 10
	(1)	(2)	(3)	(4)	(5)
$Q_{i,t-1}$	0.016***	0.020***	0.021***	0.020***	0.024***
	(24.75)	(9.40)	(9.72)	(5.57)	(3.86)
$DISA_{i,t-1}$	-1.202***	-0.722***	-0.753***	-0.690***	-0.861***
	(-7.45)	(-5.57)	(-5.82)	(-3.88)	(-3.77)
$Q_{i,t-1} \times DISA_{i,t-1}$	0.569***	0.238***	0.262***	0.281**	0.500***
	(4.90)	(2.96)	(3.25)	(2.27)	(2.87)
$ANALYST_{i,t-1}$			-0.001	-0.001	0.002
			(-0.52)	(-0.25)	(0.37)
$Q_{i,t-1} \times ANALYST_{i,t-1}$			0.003***	0.002	-0.000
			(3.31)	(1.36)	(-0.08)
$SIZE_{i,t-1}$		-0.035***	-0.034***	-0.029***	-0.029***
		(-15.94)	(-14.50)	(-9.80)	(-7.80)
$SIZE_{i,t-1} \times Q_{i,t-1}$		-0.001***	-0.003***	-0.002***	-0.002***
		(-4.45)	(-5.95)	(-4.52)	(-3.02)
$CF_{i,t}$		-0.026***	-0.082***	-0.076	-0.108
		(-2.98)	(-4.45)	(-1.52)	(-1.23)
$CF_{i,t} \times DISA_{i,t-1}$		-1.595***	-1.337**	-0.630	0.274
		(-2.98)	(-2.54)	(-0.78)	(0.25)
$CF_{i,t} \times ANALYST_{i,t-1}$			0.030***	0.036*	0.049
			(3.53)	(1.87)	(1.54)
$LEVERAGE_{i,t-1}$		-0.057***	-0.055***	-0.057***	-0.054***
		(-10.40)	(-10.05)	(-8.32)	(-6.66)
$IO_{i,t-1}$		0.008	0.002	0.012*	0.013
		(1.50)	(0.47)	(1.69)	(1.50)
$RET_{i,t+3}$		-0.002	-0.001	-0.008**	-0.007
		(-0.51)	(-0.42)	(-2.08)	(-1.46)
$INVASSET_{i,t-1}$		1.628***	1.730***	3.548***	4.126***
		(4.87)	(5.20)	(3.42)	(2.90)
Fixed Effects	Firm, Year	Firm, Year	Firm, Year	Firm, Year	Firm, Year
N	34,545	33,420	33,420	18,724	13,106
adj.R^2	0.749	0.784	0.785	0.805	0.802

Table 3: Revision around Earning Announcement Dates

This table reports the estimates from the panel regression of capital investment on Tobin's Q, revision-based disagreement, and their interaction. The investment variable is $CAPXRNDA_{i,t}$. Columns (1)–(3) use the full sample with valid revision-based disagreement measures. Column (4) restricts the sample to firms followed by at least five analysts who provide both pre- and post-earnings-announcement EPS forecasts to construct the disagreement measure. Columns (5) and (6) further separate the sample based on whether analysts' forecast revisions converge or diverge around earnings announcements. T-statistics, based on standard errors clustered at the firm level, are reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

	Full Sample			No. of Analyst ≥ 5	Converge	Diverge
	(1)	(2)	(3)	(4)	(5)	(6)
$Q_{i,t-1}$	0.014***	0.021***	0.022***	0.019**	0.021***	0.021***
	(17.62)	(7.16)	(7.34)	(2.23)	(5.90)	(3.80)
$DISA^{rev}_{i,t-1}$	-1.624***	-1.083***	-1.033***	-1.346***	-1.026***	-0.719**
	(-8.98)	(-6.38)	(-6.02)	(-4.18)	(-4.77)	(-2.03)
$Q_{i,t-1} \times DISA^{rev}_{i,t-1}$	0.881***	0.517***	0.512***	0.648***	0.504***	0.182
	(7.91)	(5.41)	(5.41)	(2.87)	(4.24)	(0.77)
$ANALYST^{rev}_{i,t-1}$			-0.007***	-0.002	-0.007***	-0.002
			(-2.93)	(-0.37)	(-2.63)	(-0.42)
$Q_{i,t-1} \times ANALYST^{rev}_{i,t-1}$			0.001	0.000	0.000	-0.000
			(0.73)	(0.14)	(0.43)	(-0.22)
$SIZE_{i,t-1}$		-0.029***	-0.028***	-0.027***	-0.030***	-0.025***
		(-11.60)	(-11.14)	(-5.29)	(-10.43)	(-6.27)
$SIZE_{i,t-1} \times Q_{i,t-1}$		-0.002***	-0.002***	-0.001*	-0.002***	-0.002**
		(-4.05)	(-4.33)	(-1.65)	(-3.34)	(-2.13)
$CF_{i,t}$		-0.007	-0.070***	0.030	-0.057***	-0.097**
		(-0.55)	(-3.58)	(0.48)	(-2.60)	(-2.32)
$CF_{i,t} \times DISA^{rev}_{i,t-1}$		-0.373	-0.690	0.548	-1.149	0.110
		(-0.60)	(-1.08)	(0.32)	(-1.49)	(0.07)
$CF_{i,t} \times ANALYST^{rev}_{i,t-1}$			0.052***	0.003	0.052***	0.054*
			(4.04)	(0.08)	(3.84)	(1.82)
$LEVERAGE_{i,t-1}$		-0.058***	-0.057***	-0.051***	-0.059***	-0.045***
		(-9.15)	(-8.99)	(-4.41)	(-8.17)	(-4.19)
$IO_{i,t-1}$		0.011*	0.011*	-0.002	0.004	0.022
		(1.67)	(1.68)	(-0.14)	(0.51)	(1.52)
$RET_{i,t+3}$		-0.003	-0.003	-0.015**	-0.004	-0.010*
		(-0.78)	(-0.88)	(-2.53)	(-0.91)	(-1.66)
$INVASSET_{i,t-1}$		2.533***	2.492***	4.779*	2.407***	3.153***
		(3.67)	(3.63)	(1.68)	(2.74)	(3.03)
Fixed Effects	Firm, Year	Firm, Year	Firm, Year	Firm, Year	Firm, Year	Firm, Year
N	18,721	18,628	18,628	4,801	12,911	4,678
adj.R^2	0.765	0.794	0.795	0.797	0.797	0.780

Table 4: Difference-in-Differences Analysis Using Regulation Fair Disclosure

This table reports covariate balance and difference-in-differences regression estimates around the implementation of Regulation Fair Disclosure in October 2000. Panel A reports the mean characteristics of treated and control firms before propensity score matching, along with the difference and its p-value. Panel B reports the same statistics after matching. Panel C reports estimates from panel regressions on the matched sample. Column (1) examines the effect of the regulation on investor disagreement ($DISA$). Column (2) examines the effect on total investment ($CAPXRNDA$) and investment-to-Q sensitivity. *Treat* equals one for firms with above-median forecast accuracy in the pre-RegFD period, and *Post* equals one for fiscal years 2001 through 2005. All regressions include firm and year fixed effects. T-statistics, based on standard errors clustered at the firm level, are reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel A Pre-matching difference in characteristics

	Treat	Control	Difference	p-value
	(1)	(2)	(1)-(2)	
$SIZE$	6.734	6.058	0.676***	0.000
Q	3.999	2.907	1.091***	0.000
$ANALYST$	2.185	1.762	0.423***	0.000
CF	0.129	0.046	0.083***	0.000
LEV	0.192	0.191	0.001	0.931
IO	0.608	0.527	0.081***	0.000
RET	-0.070	-0.162	0.092***	0.000
$INVASSET$	0.003	0.007	-0.003***	0.000

Panel B Post-matching difference in characteristics

	Treat	Control	Difference	p-value
	(1)	(2)	(1)-(2)	
$SIZE$	6.734	6.677	0.057	0.834
Q	3.999	3.703	0.296	0.522
$ANALYST$	2.185	2.179	0.007	0.959
CF	0.129	0.132	-0.003	0.923
LEV	0.192	0.180	0.013	0.558
IO	0.608	0.599	0.009	0.661
RET	-0.070	-0.063	-0.007	0.792
$INVASSET$	0.003	0.004	-0.001	0.318

Panel C DID regression

	$DISA_{i,t}$	$CAPXRNDA_{i,t}$
	(1)	(2)
$Treat \times Post \times Q_{i,t-1}$		0.0060**
		(2.33)
$Treat \times Post$	0.0012***	0.0063
	(5.03)	(0.73)
$Q_{i,t-1}$	-0.0001***	0.0254***
	(-3.82)	(4.70)
Controls	Yes	Yes
Fixed Effects	Firm, Year	Firm, Year
N	3,997	3,997
adj.R^2	0.356	0.743

Table 5: Dynamic Effects of RegFD

This table reports estimates from dynamic difference-in-differences regressions around the implementation of Regulation Fair Disclosure in October 2000. The event year is set to 2000, and the sample spans fiscal years 1995 to 2005, with fiscal year 1999 ($t = -1$) serving as the omitted base year. Column (1) reports the dynamic treatment effects on investor disagreement ($DISA$). Column (2) reports the dynamic treatment effects on total investment ($CAPXRNDA$) and investment-to-Q sensitivity. $Treat \times Post$ $(t = k)$ captures the level difference between treated and control firms at k years relative to the event, while $Treat \times Post \times Q_{i,t-1}$ $(t = k)$ captures the differential investment-to-Q sensitivity at the same horizon. All regressions are estimated on the propensity score matched sample and include firm and year fixed effects. T-statistics, based on standard errors clustered at the firm level, are reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

	$DISA_{it}$	$CAPXRNDA_{it}$
	(1)	(2)
$Treat \times Post$ $(t = -5)$	-0.0005	-0.0489***
	(-0.83)	(-2.99)
$Treat \times Post$ $(t = -4)$	0.0003	-0.0082
	(0.58)	(-0.58)
$Treat \times Post$ $(t = -3)$	0.0000	0.0013
	(0.09)	(0.12)
$Treat \times Post$ $(t = -2)$	-0.0008	0.0019
	(-1.31)	(0.22)
$Treat \times Post$ $(t = 0)$	0.0010**	0.0156*
	(2.42)	(1.66)
$Treat \times Post$ $(t = 1)$	0.0007	0.0077
	(1.64)	(0.74)
$Treat \times Post$ $(t = 2)$	0.0010**	-0.0100
	(2.51)	(-0.83)
$Treat \times Post$ $(t = 3)$	0.0013***	0.0084
	(3.48)	(0.75)
$Treat \times Post$ $(t = 4)$	0.0011***	0.0014
	(3.18)	(0.13)
$Treat \times Post$ $(t = 5)$	0.0008*	-0.0183
	(1.85)	(-1.38)
$Treat \times Post \times Q_{i,t-1}$ $(t = -5)$		0.0072
		(1.59)
$Treat \times Post \times Q_{i,t-1}$ $(t = -4)$		0.0019
		(0.54)
$Treat \times Post \times Q_{i,t-1}$ $(t = -3)$		0.0025
		(0.91)
$Treat \times Post \times Q_{i,t-1}$ $(t = -2)$		0.0018
		(0.92)
$Treat \times Post \times Q_{i,t-1}$ $(t = 0)$		-0.0025
		(-1.43)
$Treat \times Post \times Q_{i,t-1}$ $(t = 1)$		0.0026
		(1.00)
$Treat \times Post \times Q_{i,t-1}$ $(t = 2)$		0.0137***
		(3.06)
$Treat \times Post \times Q_{i,t-1}$ $(t = 3)$		0.0065**
		(2.02)
$Treat \times Post \times Q_{i,t-1}$ $(t = 4)$		0.0072**
		(2.16)
$Treat \times Post \times Q_{i,t-1}$ $(t = 5)$		0.0179***
		(3.90)
Controls	Yes	Yes
Fixed Effects	Firm, Year	Firm, Year
N	4,444	4,444
adj.R^2	0.342	0.747

Table 6: IV Regression Using Reg FD

This table reports instrumental variable regression estimates using Regulation Fair Disclosure as the source of exogenous variation, estimated on the propensity score matched sample. The endogenous variables are investor disagreement ($DISA$) and its interaction with Tobin's Q ($Q \times DISA$). The excluded instruments are $Treat \times Post$ and $Treat \times Post \times Q_{i,t-1}$. Columns (1) and (2) report the first-stage regressions for $DISA$ and $Q \times DISA$, respectively. Column (3) reports the second-stage regression, where the dependent variable is total investment ($CAPXRNDA$). Excluded-IV F reports the Kleibergen-Paap Wald F-statistic for weak instrument diagnostics. All regressions include firm and year fixed effects. T-statistics, based on standard errors clustered at the firm level, are reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

	First-stage		Second-stage
	$DISA_{i,t}$	$Q_{i,t-1} \times DISA_{i,t}$	$CAPXRNDA_{i,t}$
	(1)	(2)	(3)
$Treat \times Post$	0.0013***	0.0027***	
	(4.74)	(4.66)	
$Treat \times Post \times Q_{i,t-1}$	-0.0001	0.0002	
	(-0.97)	(1.27)	
$Q_{i,t-1} \times DISA_{i,t}$ (IV)			23.0637*
			(1.92)
$DISA_{i,t}$ (IV)			-41.8942
			(-1.47)
$Q_{i,t-1}$	-0.0004***	0.0010	-0.0117
	(-2.79)	(1.42)	(-0.51)
Excluded-IV F	13.24	13.96	
Controls	Yes	Yes	Yes
Fixed Effects	Firm, Year	Firm, Year	Firm, Year
N	3,997	3,997	3,997

Table 7: Cross-sectional Analysis: CAPEX v.s. R&D

This table reports estimates from panel regressions of capital investment on Tobin's Q, investor disagreement, and their interaction. The investment variable is capital expenditure scaled by lagged total assets ($CAPXA$) in Columns (1) and (3), and R&D expenses scaled by lagged total assets ($RNDA$) in Columns (2) and (4). Columns (1) to (2) use the sample based on the availability of each dependent variable, while Columns (3) to (4) restrict the sample to firms with both *CAPEX* and *R&D* investments available. T-statistics, based on standard errors clustered at the firm level, are reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

	Full sample		Restricted sample	
Dependent Variable is	$CAPXA_{i,t}$	$RNDA_{i,t}$	$CAPXA_{i,t}$	$RNDA_{i,t}$
	(1)	(2)	(3)	(4)
$Q_{i,t-1}$	0.008***	0.013***	0.009***	0.013***
	(5.90)	(8.45)	(6.21)	(8.45)
$DISA_{i,t-1}$	-0.664***	-0.247***	-0.490***	-0.245***
	(-9.03)	(-2.87)	(-6.29)	(-2.83)
$Q_{i,t-1} \times DISA_{i,t-1}$	0.103**	0.202***	0.035	0.204***
	(2.50)	(3.26)	(0.91)	(3.29)
$ANALYST_{i,t-1}$	-0.000	-0.000	-0.001	-0.000
	(-0.20)	(-0.27)	(-0.53)	(-0.16)
$Q_{i,t-1} \times ANALYST_{i,t-1}$	0.000	0.002***	0.001**	0.002***
	(1.02)	(2.90)	(1.99)	(2.94)
$SIZE_{i,t-1}$	-0.014***	-0.022***	-0.011***	-0.022***
	(-11.10)	(-12.90)	(-8.19)	(-12.95)
$SIZE_{i,t-1} \times Q_{i,t-1}$	-0.001***	-0.002***	-0.001***	-0.002***
	(-2.76)	(-5.41)	(-3.86)	(-5.41)
$CF_{i,t}$	0.070***	-0.129***	0.055***	-0.129***
	(6.95)	(-9.38)	(5.30)	(-9.41)
$CF_{i,t} \times DISA_{i,t-1}$	-1.123***	0.173	-1.359***	0.224
	(-3.84)	(0.46)	(-4.53)	(0.60)
$CF_{i,t} \times ANALYST_{i,t-1}$	0.007	0.024***	0.005	0.023***
	(1.58)	(3.73)	(1.09)	(3.65)
$LEVERAGE_{i,t-1}$	-0.041***	-0.015***	-0.037***	-0.015***
	(-12.54)	(-3.99)	(-10.36)	(-4.06)
$IO_{i,t-1}$	0.008***	-0.002	0.005	-0.003
	(2.81)	(-0.71)	(1.46)	(-0.81)
$RET_{i,t+3}$	-0.016***	0.013***	-0.014***	0.013***
	(-9.60)	(6.15)	(-7.57)	(6.12)
$INVASSET_{i,t-1}$	0.663***	0.748***	0.642***	0.762***
	(3.59)	(3.40)	(3.32)	(3.47)
Fixed Effects	Firm, Year	Firm, Year	Firm, Year	Firm, Year
N	51,209	33,645	33,420	33,420
adj.R^2	0.692	0.882	0.651	0.883

Table 8: Cross-sectional Analysis: Growth v.s. Value Firms

This table reports estimates from panel regressions examining whether the impact of investor disagreement on the investment-to-Q sensitivity differs between value and growth firms. Firms are classified into growth (Columns (1)–(3)) and value (Columns (4)–(6)) subsamples based on whether their book-to-market (BM) ratios are below or above the sample median, respectively, in each fiscal year. For each subsample, we estimate three regressions, where the dependent variable is *CAPXRNDA*, *CAPXA*, and *RNDA*, respectively. T-statistics, based on standard errors clustered at the firm level, are reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

	Growth Firms (BM lower than sample median)			Value Firms (BM higher than sample median)		
	$CAPXRNDA_{i,t}$	$CAPXA_{i,t}$	$RNDA_{i,t}$	$CAPXRNDA_{i,t}$	$CAPXA_{i,t}$	$RNDA_{i,t}$
	(1)	(2)	(3)	(4)	(5)	(6)
$Q_{i,t-1}$	0.019***	0.008***	0.012***	0.064***	0.029***	0.025***
	(7.63)	(5.58)	(6.80)	(5.42)	(3.59)	(3.63)
$DISA_{i,t-1}$	-1.183***	-1.115***	-0.416**	-0.081	-0.672***	0.244
	(-4.15)	(-5.92)	(-2.23)	(-0.29)	(-4.21)	(1.29)
$Q_{i,t-1} \times DISA_{i,t-1}$	0.331***	0.167***	0.226***	-0.110	0.391**	-0.239
	(3.19)	(2.83)	(2.92)	(-0.38)	(2.54)	(-1.22)
$ANALYST_{i,t-1}$	-0.000	0.001	0.001	-0.006	-0.004	0.000
	(-0.06)	(0.30)	(0.27)	(-1.57)	(-1.50)	(0.14)
$Q_{i,t-1} \times ANALYST_{i,t-1}$	0.003***	0.001	0.002***	0.005	0.003	-0.000
	(2.76)	(0.99)	(2.58)	(1.64)	(1.31)	(-0.06)
$SIZE_{i,t-1}$	-0.039***	-0.013***	-0.027***	-0.013***	-0.006***	-0.011***
	(-11.25)	(-7.34)	(-11.03)	(-3.87)	(-2.83)	(-5.73)
$SIZE_{i,t-1} \times Q_{i,t-1}$	-0.002***	-0.001***	-0.002***	-0.008***	-0.002**	-0.003***
	(-4.67)	(-3.16)	(-4.31)	(-4.43)	(-2.00)	(-2.94)
$CF_{i,t}$	-0.100***	0.063***	-0.147***	-0.048*	0.061***	-0.072***
	(-4.16)	(4.83)	(-8.20)	(-1.77)	(3.81)	(-3.42)
$CF_{i,t} \times DISA_{i,t-1}$	-1.427**	-1.233***	0.019	-2.148***	-0.859**	-1.187**
	(-1.98)	(-2.85)	(0.03)	(-2.74)	(-2.12)	(-2.33)
$CF_{i,t} \times ANALYST_{i,t-1}$	0.026**	-0.001	0.025***	0.044***	0.018**	0.025***
	(2.35)	(-0.11)	(3.06)	(3.66)	(2.40)	(2.76)
$LEVERAGE_{i,t-1}$	-0.056***	-0.031***	-0.026***	-0.067***	-0.055***	-0.009**
	(-8.00)	(-7.77)	(-4.86)	(-8.89)	(-10.63)	(-2.54)
$IO_{i,t-1}$	0.008	0.004	0.005	-0.011	0.005	-0.008**
	(1.07)	(0.89)	(0.91)	(-1.60)	(1.15)	(-2.53)
$RET_{i,t+3}$	0.007	-0.012***	0.018***	-0.010***	-0.012***	0.003
	(1.39)	(-4.87)	(5.09)	(-2.86)	(-5.05)	(1.43)
$INVASSET_{i,t-1}$	1.792***	0.443**	0.898***	2.003***	0.785*	0.878*
	(4.57)	(2.03)	(3.39)	(2.74)	(1.65)	(1.77)
Fixed Effects	Firm, Year	Firm, Year	Firm, Year	Firm, Year	Firm, Year	Firm, Year
N	18,548	25,280	18,656	13,760	24,334	13,875
adj.R^2	0.788	0.716	0.875	0.773	0.713	0.909

Table 9: Cross-sectional Analysis: Less v.s. More Analyst Coverage

This table reports estimates from panel regressions examining whether the impact of investor disagreement on the investment-to-Q sensitivity differs between firms with low and high analyst coverage. Firm are classified into low and high coverage groups based on the median value of number of analyst coverage ($ANALYST_{i,t-1}$). For each subsample, we estimate three regressions, where the dependent variable is *CAPXRNDA*, *CAPXA*, and *RNDA*, respectively. T-statistics, based on standard errors clustered at the firm level, are reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

	Low Analyst Coverage			High Analyst Coverage		
	$CAPXRNDA_{i,t}$	$CAPXA_{i,t}$	$RNDA_{i,t}$	$CAPXRNDA_{i,t}$	$CAPXA_{i,t}$	$RNDA_{i,t}$
	(1)	(2)	(3)	(4)	(5)	(6)
$Q_{i,t-1}$	0.020***	0.006**	0.013***	0.023***	0.009***	0.013***
	(4.09)	(2.49)	(4.21)	(6.45)	(4.05)	(3.45)
$DISA_{i,t-1}$	-0.710***	-0.467***	-0.319***	-1.024***	-1.052***	-0.454***
	(-3.91)	(-3.80)	(-3.05)	(-3.79)	(-4.49)	(-3.24)
$Q_{i,t-1}\times DISA_{i,t-1}$	0.208**	0.059	0.168**	0.554***	0.302**	0.494***
	(2.02)	(1.04)	(2.37)	(2.98)	(2.15)	(3.92)
$ANALYST_{i,t-1}$	0.006	0.004**	0.002	0.001	-0.000	-0.002
	(1.67)	(2.57)	(0.69)	(0.20)	(-0.08)	(-0.54)
$Q_{i,t-1}\times ANALYST_{i,t-1}$	0.001	-0.001	0.002	0.001	0.000	0.001
	(0.94)	(-1.46)	(1.66)	(0.37)	(0.01)	(0.77)
$SIZE_{i,t-1}$	-0.043***	-0.015***	-0.028***	-0.029***	-0.013***	-0.018***
	(-8.96)	(-6.67)	(-8.99)	(-9.70)	(-7.26)	(-7.64)
$SIZE_{i,t-1}\times Q_{i,t-1}$	-0.002**	0.000	-0.002***	-0.002***	-0.001**	-0.001***
	(-2.25)	(0.17)	(-3.12)	(-4.16)	(-2.49)	(-3.42)
$CF_{i,t}$	-0.033	0.092***	-0.099***	-0.076	0.089**	-0.159***
	(-1.06)	(5.47)	(-4.30)	(-1.14)	(2.19)	(-3.37)
$CF_{i,t}\times DISA_{i,t-1}$	-1.968**	-1.063**	-0.330	0.018	-1.265*	1.156*
	(-2.61)	(-2.63)	(-0.64)	(0.02)	(-1.83)	(1.72)
$CF_{i,t}\times ANALYST_{i,t-1}$	-0.008	-0.015	0.007	0.037	0.005	0.039**
	(-0.41)	(-1.58)	(0.55)	(1.43)	(0.33)	(2.20)
$LEVERAGE_{i,t-1}$	-0.048***	-0.038***	-0.010*	-0.059***	-0.042***	-0.019***
	(-4.97)	(-6.58)	(-1.74)	(-6.86)	(-7.07)	(-3.76)
$IO_{i,t-1}$	0.000	0.009**	-0.002	0.009	0.005	0.003
	(0.04)	(2.38)	(-0.44)	(1.17)	(1.02)	(0.58)
$RET_{i,t+3}$	0.001	-0.013***	0.014***	-0.009	-0.022***	0.011***
	(0.27)	(-5.20)	(4.34)	(-1.62)	(-5.75)	(3.25)
$INVASSET_{i,t-1}$	1.332***	0.716***	0.354	3.685***	1.320**	1.829**
	(2.76)	(3.34)	(1.11)	(3.11)	(2.52)	(2.32)
Fixed Effects	Firm, Year	Firm, Year	Firm, Year	Firm, Year	Firm, Year	Firm, Year
N	15,540	24,748	15,641	16,933	25,076	17,059
adj.R^2	0.789	0.671	0.885	0.795	0.723	0.890

Table 10: Cross-sectional Analysis: RE v.s. DO

This table reports estimates from panel regressions examining whether the effect of investor disagreement on investment-to-Q sensitivity differs between firms in which investors condition less strongly on prices (DO) and firms in which investors condition strongly on prices (RE). Following Banerjee (2011), firms are classified into the DO subsample in Columns (1)–(3) and the RE subsample in Columns (4)–(6) according to whether the covariance between realized return variance and trading volume, $Cov(V,T)_{i,t-1}$, is below or above the sample median in each fiscal year, respectively. For each subsample, we estimate three regressions in which the dependent variables are *CAPXRNDA*, *CAPXA*, and *RNDA*, respectively. T-statistics based on standard errors clustered at the firm level are reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

	DO			RE		
	$CAPXRNDA_{i,t}$	$CAPXA_{i,t}$	$RNDA_{i,t}$	$CAPXRNDA_{i,t}$	$CAPXA_{i,t}$	$RNDA_{i,t}$
	(1)	(2)	(3)	(4)	(5)	(6)
$Q_{i,t-1}$	0.026***	0.008***	0.017***	0.020***	0.008***	0.013***
	(7.37)	(3.50)	(6.77)	(6.54)	(4.14)	(6.14)
$DISA_{i,t-1}$	-0.414**	-0.633***	-0.141	-1.178***	-0.581***	-0.509***
	(-2.18)	(-5.49)	(-1.11)	(-5.47)	(-4.23)	(-3.67)
$Q_{i,t-1}\times DISA_{i,t-1}$	0.086	0.109	0.094	0.548***	0.028	0.436***
	(0.67)	(1.61)	(0.98)	(4.00)	(0.40)	(4.15)
$ANALYST_{i,t-1}$	-0.002	-0.002	0.001	0.002	0.001	0.000
	(-0.62)	(-1.15)	(0.41)	(0.52)	(0.34)	(0.14)
$Q_{i,t-1}\times ANALYST_{i,t-1}$	0.002*	0.000	0.001	0.002	0.001	0.001
	(1.90)	(0.46)	(1.45)	(1.60)	(1.09)	(1.51)
$SIZE_{i,t-1}$	-0.028***	-0.011***	-0.019***	-0.036***	-0.012***	-0.023***
	(-9.29)	(-7.12)	(-8.58)	(-11.83)	(-6.32)	(-10.83)
$SIZE_{i,t-1}\times Q_{i,t-1}$	-0.003***	-0.001*	-0.002***	-0.002***	-0.001**	-0.002***
	(-4.52)	(-1.72)	(-4.37)	(-3.91)	(-2.49)	(-3.76)
$CF_{i,t}$	-0.081***	0.049***	-0.116***	-0.080***	0.060***	-0.135***
	(-2.67)	(3.09)	(-5.21)	(-2.92)	(3.78)	(-6.68)
$CF_{i,t}\times DISA_{i,t-1}$	-2.426**	-0.874	-0.072	-0.459	-0.993*	0.606
	(-2.41)	(-1.61)	(-0.11)	(-0.53)	(-1.94)	(0.98)
$CF_{i,t}\times ANALYST_{i,t-1}$	0.020	0.023***	0.010	0.034***	0.004	0.029***
	(1.43)	(2.90)	(0.96)	(2.86)	(0.66)	(3.36)
$LEVERAGE_{i,t-1}$	-0.049***	-0.038***	-0.014***	-0.068***	-0.044***	-0.020***
	(-6.48)	(-8.90)	(-2.67)	(-9.58)	(-8.95)	(-4.34)
$IO_{i,t-1}$	0.001	0.007*	-0.006	0.000	0.006	-0.005
	(0.14)	(1.92)	(-1.50)	(0.04)	(1.07)	(-0.95)
$RET_{i,t+3}$	-0.001	-0.016***	0.013***	-0.000	-0.013***	0.014***
	(-0.32)	(-7.02)	(4.42)	(-0.10)	(-4.90)	(4.85)
$INVASSET_{i,t-1}$	2.019***	0.854***	0.657**	1.143**	0.423	0.568*
	(4.72)	(3.29)	(2.50)	(2.16)	(1.41)	(1.65)
Fixed Effects	Firm, Year	Firm, Year	Firm, Year	Firm, Year	Firm, Year	Firm, Year
N	13,729	23,080	13,828	16,859	16,859	16,978
adj.R^2	0.809	0.716	0.901	0.766	0.636	0.869

Table 11: Cross-sectional Analysis: Less v.s. More Financial Constraints

This table reports estimates from panel regressions examining whether the impact of investor disagreement on investment-to-Q sensitivity differs between firms facing low versus high financial constraints. Financial constraints (FC) are defined following Edmans et al. (2017) as the difference between capital expenditure and cash flow, scaled by total capital expenditure. Firms are classified into more financially constrained (Columns (1)–(3)) and less financially constrained (Columns (4)–(6)) subsamples based on whether their FC values are below or above the sample median in each fiscal year, respectively. For each subsample, we estimate three regressions, where the dependent variable is $CAPXRNDA$, $CAPXA$, and $RNDA$, respectively. T-statistics, based on standard errors clustered at the firm level, are reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

	More Financial Constraints			Less Financial Constraints		
	$CAPXRNDA_{i,t}$	$CAPXA_{i,t}$	$RNDA_{i,t}$	$CAPXRNDA_{i,t}$	$CAPXA_{i,t}$	$RNDA_{i,t}$
	(1)	(2)	(3)	(4)	(5)	(6)
$Q_{i,t-1}$	0.014***	0.010***	0.003	0.018***	0.004**	0.013***
	(4.49)	(5.25)	(1.27)	(6.59)	(2.48)	(6.17)
$DISA_{i,t-1}$	-0.251	-0.471***	0.009	-0.855***	-0.774***	-0.230**
	(-0.71)	(-2.87)	(0.04)	(-5.86)	(-8.49)	(-2.44)
$Q_{i,t-1}\times DISA_{i,t-1}$	0.013	0.025	-0.035	0.246***	0.094**	0.180***
	(0.05)	(0.19)	(-0.19)	(2.97)	(2.11)	(2.82)
$ANALYST_{i,t-1}$	-0.002	-0.000	-0.001	0.007**	0.002	0.005**
	(-0.96)	(-0.03)	(-0.49)	(2.21)	(1.21)	(2.30)
$Q_{i,t-1}\times ANALYST_{i,t-1}$	0.003***	0.001	0.002*	0.002	-0.001	0.001*
	(2.76)	(1.46)	(1.84)	(1.52)	(-0.99)	(1.79)
$SIZE_{i,t-1}$	-0.015***	-0.008***	-0.008***	-0.050***	-0.019***	-0.034***
	(-7.48)	(-6.53)	(-6.72)	(-15.15)	(-9.80)	(-13.86)
$SIZE_{i,t-1}\times Q_{i,t-1}$	-0.002***	-0.001***	-0.000	-0.002***	0.000	-0.001***
	(-3.83)	(-3.66)	(-1.02)	(-2.77)	(0.80)	(-3.11)
$CF_{i,t}$	0.046	0.083***	-0.026	-0.056**	0.095***	-0.119***
	(1.30)	(5.06)	(-1.01)	(-2.56)	(7.36)	(-7.28)
$CF_{i,t}\times DISA_{i,t-1}$	-1.561	-0.397	0.803	-1.915***	-1.614***	-0.143
	(-1.12)	(-0.52)	(0.95)	(-3.49)	(-4.67)	(-0.36)
$CF_{i,t}\times ANALYST_{i,t-1}$	-0.002	0.008	-0.009	0.022**	-0.003	0.023***
	(-0.12)	(1.00)	(-0.65)	(2.22)	(-0.48)	(3.06)
$LEVERAGE_{i,t-1}$	-0.037***	-0.029***	-0.010***	-0.072***	-0.058***	-0.021***
	(-7.11)	(-9.21)	(-3.00)	(-9.42)	(-11.86)	(-3.77)
$IO_{i,t-1}$	0.001	0.003	0.002	0.015**	0.015***	0.005
	(0.22)	(1.01)	(0.85)	(2.16)	(3.31)	(1.09)
$RET_{i,t+3}$	-0.011***	-0.011***	0.002	0.001	-0.017***	0.015***
	(-3.79)	(-6.31)	(1.40)	(0.23)	(-6.72)	(4.83)
$INVASSET_{i,t-1}$	0.395	0.117	0.122	1.736***	0.618**	0.800***
	(1.00)	(0.46)	(0.46)	(4.13)	(2.50)	(2.89)
Fixed Effects	Firm, Year	Firm, Year	Firm, Year	Firm, Year	Firm, Year	Firm, Year
N	14,115	25,392	14,229	18,367	24,229	18,440
adj.R^2	0.713	0.659	0.821	0.756	0.709	0.865

Table 12: Addressing Valuation Uncertainty

This table reports the estimates from the panel regression examining the impacts of valuation uncertainty (VU)-driven disagreement and the residual disagreement on the investment-to-Q sensitivity. The investment variable is *CAPXRNDA*. $\widehat{DISA}_{it-1}$ represents the component of disagreement driven by valuation uncertainty (VU), obtained from regressing investor disagreement on one of two VU proxies, earnings volatility (Columns (1)–(3)) or Loughran and McDonald (2024)'s firm complexity (Columns (4)–(6)). $DISA^{\perp}_{it-1}$ is the corresponding residual disagreement component orthogonal to these proxies. T-statistics, based on standard errors clustered at the firm level, are reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

	VU Proxy is Earnings Volatility			VU Proxy is Firm Complexity		
	(1)	(2)	(3)	(4)	(5)	(6)
$Q_{i,t-1}$	0.022***	0.025***	0.022***	0.018***	0.019***	0.018***
	(7.28)	(8.72)	(7.54)	(6.77)	(7.73)	(7.03)
$\widehat{DISA}_{i,t-1}$	-3.918***		-3.815***	-6.864**		-6.923**
	(-5.38)		(-5.13)	(-2.06)		(-2.07)
$Q_{i,t-1}\times\widehat{DISA}_{i,t-1}$	0.242		0.461**	0.152		0.352
	(1.28)		(2.07)	(0.73)		(1.60)
$DISA^{\perp}_{i,t-1}$		-1.025***	-0.998***		-1.933***	-1.981***
		(-5.51)	(-5.09)		(-6.30)	(-6.60)
$Q_{i,t-1}\times DISA^{\perp}_{i,t-1}$		0.575***	0.582***		0.681***	0.721***
		(4.29)	(3.85)		(4.56)	(4.92)
$ANALYST_{i,t-1}$	-0.000	-0.001	-0.001	-0.001	-0.002	-0.002
	(-0.12)	(-0.28)	(-0.37)	(-0.44)	(-0.77)	(-0.83)
$Q_{i,t-1}\times ANALYST_{i,t-1}$	0.002**	0.002**	0.002**	0.002**	0.002**	0.003**
	(1.99)	(2.16)	(2.24)	(2.04)	(2.50)	(2.53)
$SIZE_{i,t-1}$	-0.032***	-0.030***	-0.031***	-0.035***	-0.035***	-0.035***
	(-12.34)	(-11.71)	(-12.06)	(-12.54)	(-12.54)	(-12.37)
$SIZE_{i,t-1}\times Q_{i,t-1}$	-0.002***	-0.003***	-0.002***	-0.002***	-0.002***	-0.002***
	(-4.70)	(-5.34)	(-4.93)	(-3.24)	(-3.76)	(-3.65)
$CF_{i,t}$	-0.065***	-0.069***	-0.059**	-0.076***	-0.090***	-0.072***
	(-2.73)	(-2.98)	(-2.52)	(-3.22)	(-4.24)	(-3.02)
$CF_{i,t}\times\widehat{DISA}_{i,t-1}$	-0.689		-1.305	-3.910**		-3.748*
	(-0.53)		(-1.02)	(-2.02)		(-1.92)
$CF_{i,t}\times DISA^{\perp}_{i,t-1}$		-1.597*	-1.668*		-0.102	-0.202
		(-1.85)	(-1.96)		(-0.09)	(-0.19)
$CF_{i,t}\times ANALYST_{i,t-1}$	0.028***	0.028***	0.026**	0.021**	0.020**	0.018*
	(2.58)	(2.62)	(2.42)	(2.08)	(2.00)	(1.80)
$LEVERAGE_{i,t-1}$	-0.052***	-0.053***	-0.052***	-0.050***	-0.050***	-0.049***
	(-8.80)	(-8.84)	(-8.59)	(-7.46)	(-7.48)	(-7.30)
$IO_{i,t-1}$	-0.000	-0.001	-0.002	0.010*	0.009	0.009
	(-0.00)	(-0.09)	(-0.26)	(1.69)	(1.61)	(1.46)
$RET_{i,t+3}$	-0.008**	-0.007**	-0.007**	0.001	0.002	0.002
	(-2.44)	(-2.09)	(-2.06)	(0.23)	(0.68)	(0.61)
$INVASSET_{i,t-1}$	1.454**	1.328**	1.256**	3.121***	2.983***	2.982***
	(2.53)	(2.36)	(2.21)	(6.79)	(6.49)	(6.50)
Fixed Effects	Firm, Year	Firm, Year	Firm, Year	Firm, Year	Firm, Year	Firm, Year
N	25,137	25,137	25,137	21,588	21,588	21,588
adj.R^2	0.794	0.794	0.795	0.811	0.812	0.812

Table 13: Response to Peer's Q

This table reports the estimates from the panel regression examining the impact of investor disagreement on the sensitivity of investment to peer's Q. The investment variable is *CAPXRNDA*. Firms are classified into the same industry using one of two industry classification schemes, Hoberg and Phillips (2016) text-based Industrial Code (Columns (1)–(3)) or 2-digit Standard Industrial Code (SIC) (Columns (4)–(6)). When aggregating peer Q, we consider three alternatives: the equally weighted mean, the value-weighted mean based on firm size, and the industry median. T-statistics, based on standard errors clustered at the firm level, are reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

	Hoberg and Phillips (2016) FIC-100 Industry			SIC 2-digit Industry		
	EW Average	VW Average	Median	EW Average	VW Average	Median
	(1)	(2)	(3)	(4)	(5)	(6)
$PeerQ_{i,t-1}$	0.027***	0.027***	0.032***	0.044***	0.047***	0.063***
	(5.13)	(5.09)	(4.59)	(9.61)	(9.66)	(9.98)
$DISA_{i,t-1}$	-1.359***	-1.402***	-1.084**	-0.232	-0.215	-0.101
	(-2.86)	(-2.93)	(-2.20)	(-1.28)	(-1.19)	(-0.52)
$PeerQ_{i,t-1} \times DISA_{i,t-1}$	0.015	0.034	-0.109	-0.168*	-0.183**	-0.280**
	(0.08)	(0.18)	(-0.47)	(-1.95)	(-2.04)	(-2.43)
$ANALYST_{i,t-1}$	-0.015***	-0.016***	-0.018***	-0.010***	-0.011***	-0.014***
	(-3.37)	(-3.46)	(-4.06)	(-3.06)	(-3.11)	(-3.87)
$PeerQ_{i,t-1} \times ANALYST_{i,t-1}$	0.007***	0.008***	0.010***	0.007***	0.008***	0.011***
	(3.91)	(3.99)	(4.49)	(5.13)	(5.13)	(5.69)
$SIZE_{i,t-1}$	-0.041***	-0.041***	-0.040***	-0.032***	-0.032***	-0.029***
	(-10.54)	(-10.46)	(-10.20)	(-11.98)	(-11.67)	(-10.62)
$SIZE_{i,t-1} \times PeerQ_{i,t-1}$	-0.006***	-0.006***	-0.007***	-0.007***	-0.008***	-0.010***
	(-5.59)	(-5.61)	(-5.69)	(-9.58)	(-9.57)	(-10.37)
$CF_{i,t}$	-0.084***	-0.084***	-0.086***	-0.070***	-0.071***	-0.071***
	(-3.01)	(-3.00)	(-3.09)	(-3.62)	(-3.63)	(-3.65)
$CF_{i,t} \times DISA_{i,t-1}$	-1.486	-1.457	-1.654	-2.000***	-2.032***	-2.116***
	(-1.03)	(-1.01)	(-1.12)	(-3.40)	(-3.44)	(-3.54)
$CF_{i,t} \times ANALYST_{i,t-1}$	0.038***	0.038***	0.039***	0.039***	0.039***	0.039***
	(2.95)	(2.95)	(3.04)	(4.33)	(4.35)	(4.38)
$LEVERAGE_{i,t-1}$	-0.053***	-0.052***	-0.052***	-0.058***	-0.058***	-0.057***
	(-6.54)	(-6.53)	(-6.50)	(-10.36)	(-10.34)	(-10.28)
$IO_{i,t-1}$	0.019***	0.020***	0.019***	0.008	0.008	0.008
	(2.99)	(3.01)	(2.87)	(1.54)	(1.53)	(1.47)
$RET_{i,t+3}$	-0.028***	-0.029***	-0.029***	-0.022***	-0.022***	-0.022***
	(-7.43)	(-7.44)	(-7.56)	(-7.61)	(-7.59)	(-7.70)
$INVASSET_{i,t-1}$	3.666***	3.646***	3.662***	2.545***	2.570***	2.541***
	(7.32)	(7.28)	(7.27)	(7.72)	(7.79)	(7.71)
Fixed Effects	Firm, Year	Firm, Year	Firm, Year	Firm, Year	Firm, Year	Firm, Year
N	17,186	17,186	17,186	33,398	33,398	33,398
adj.R^2	0.785	0.785	0.785	0.775	0.775	0.775

Table 14: Alternative Disagreement Measures

This table reports the estimates from the panel regression examining the alternative disagreements on the investment-to-Q sensitivity. These alternative measures are as follows, 13F institutional investors Buyer-Seller Disagreement ($DISA^{BSD}$), 13F institutional trading dispersion ($DISA^{TD}$), analyst dispersion scaled by mean forecast ($DISA^{Mean}$), analyst dispersion scaled by assets per share ($DISA^{ATS}$), and standardized unexplained trading volume ($DISA^{SUV}$), from Column (1) to (5). Control variables are included in the regression but omitted in presentation for brevity. T-statistics, based on standard errors clustered at the firm level, are reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

	13F Buyer-Seller Disagreement	13F Institutional Trading Dispersion	AD scaled by mean forecast	AD scaled by Assets Per Share	Standardized Unexplained Volume
	(1)	(2)	(3)	(4)	(5)
$Q_{i,t-1}$	0.021***	0.031***	0.024***	0.020***	0.022***
	(9.79)	(9.92)	(9.77)	(7.74)	(6.04)
$DISA^{BSD}_{i,t-1}$	-0.002***				
	(-2.94)				
$Q_{i,t-1}\times DISA^{BSD}_{i,t-1}$	0.001***				
	(3.15)				
$DISA^{TD}_{i,t-1}$		-0.017***			
		(-5.43)			
$Q_{i,t-1}\times DISA^{TD}_{i,t-1}$		0.005***			
		(3.94)			
$DISA^{Mean}_{i,t-1}$			-0.005***		
			(-6.50)		
$Q_{i,t-1}\times DISA^{Mean}_{i,t-1}$			0.001**		
			(2.28)		
$DISA^{ATS}_{i,t-1}$				-0.000	
				(-0.35)	
$Q_{i,t-1}\times DISA^{ATS}_{i,t-1}$				0.001**	
				(2.04)	
$DISA^{SUV}_{i,t-1}$					0.000
					(0.16)
$Q_{i,t-1}\times DISA^{SUV}_{i,t-1}$					-0.000
					(-0.14)
Fixed Effects	Firm, Year	Firm, Year	Firm, Year	Firm, Year	Firm, Year
Controls	Yes	Yes	Yes	Yes	Yes
N	33,405	33,405	30,823	30,873	22,762
adj.R^2	0.784	0.785	0.794	0.795	0.772

Table 15: Single Sorts by Lagged Disagreement

This table presents univariate portfolio-sorting results for the relation between investor disagreement, *DISA*, and proxies for private information. The four proxies are price nonsynchronicity (*Nonsync*), the probability of informed trading developed by Easley et al. (1996) (*PIN*), the generalised probability of informed trading developed by Duarte et al. (2020) (*GPIN*), and the private information share developed by Brogaard et al. (2022) (*PrivateInfo*). In each fiscal year, stocks are sorted into five quintile portfolios based on $DISA_{i,t-1}$. We calculate the mean value of each private information proxy for each quintile and then report the time-series average of these annual portfolio means. HML is the difference between the high- and low-disagreement portfolios, and the corresponding time-series t-statistic, calculated using Newey and West (1987) standard errors with five lags, is reported in parentheses. *N_Years* denotes the number of fiscal years included in each test. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

	Low	Q2	Q3	Q4	High	HML	T	N_Years
$Nonsync_{i,t}$	0.811	0.769	0.812	0.910	1.052	0.241***	(3.79)	49
$PIN_{i,t}$	0.155	0.150	0.159	0.168	0.180	0.025***	(4.85)	18
$GPIN_{i,t}$	0.066	0.069	0.074	0.077	0.086	0.021***	(9.65)	22
$PrivateInfo_{i,t}$	0.262	0.253	0.261	0.261	0.273	0.011*	(1.82)	46

Figures

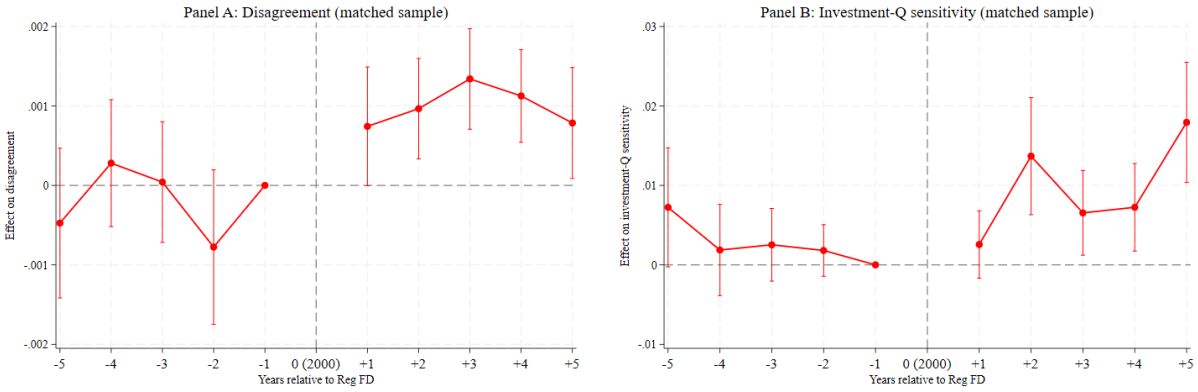


Figure 1: Parallel Trends for Disagreement and Investment-to-Q Sensitivity around RegFD

This figure plots the coefficients for the parallel trends tests around RegFD, with the estimation results are presented in Table 5. The left panel plots the coefficients on the interactions between *Treat* and event-time indicators for years relative to Reg FD, using $t = -1$ as the omitted reference period. These coefficients capture the dynamic effect of Reg FD on investor disagreement. The right panel plots the coefficients on the interactions between *Treat*, Q_{it-1}, and the same event-time indicators. These coefficients capture changes in investment-to-Q sensitivity around Reg FD.

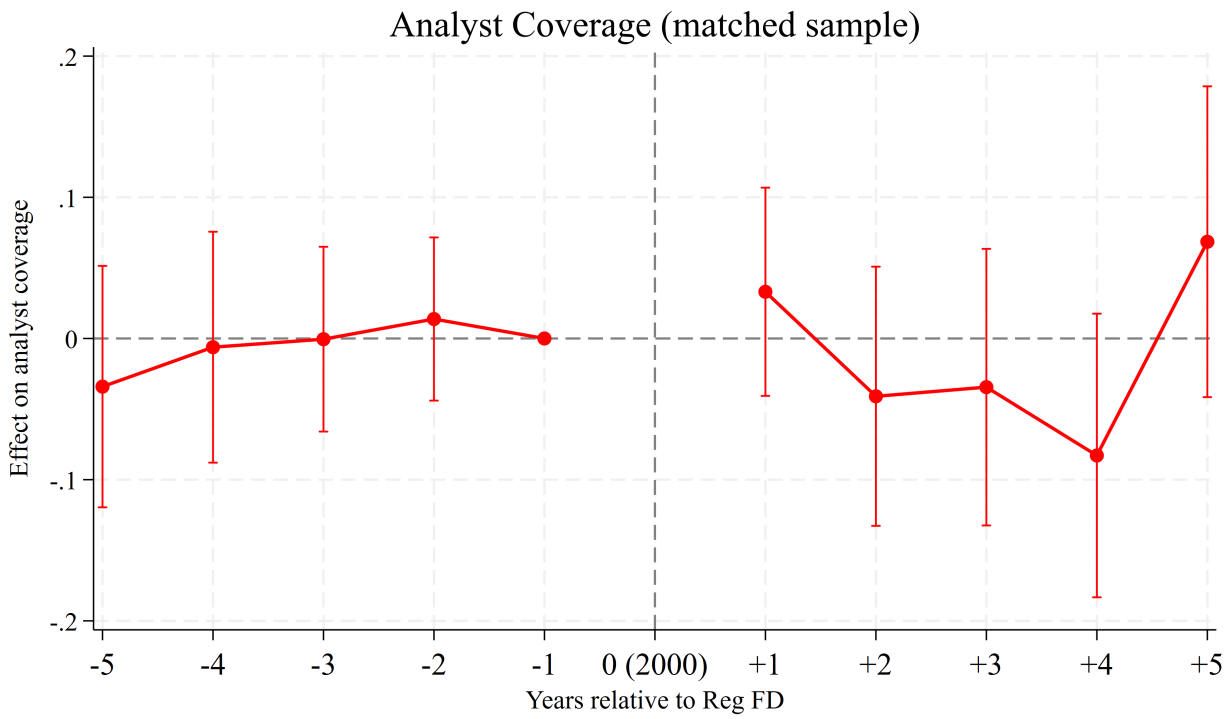


Figure 2: Parallel Trends for Analyst Coverage around RegFD

This figure plots the coefficients for the parallel trends test for analyst coverage ($ANALYST_{it}$) around RegFD. It plots the coefficients on the interactions between $Treat$ and event-time indicators for years relative to Reg FD, using $t = -1$ as the omitted reference period. These coefficients capture the dynamic effect of Reg FD on $ANALYST_{it}$.

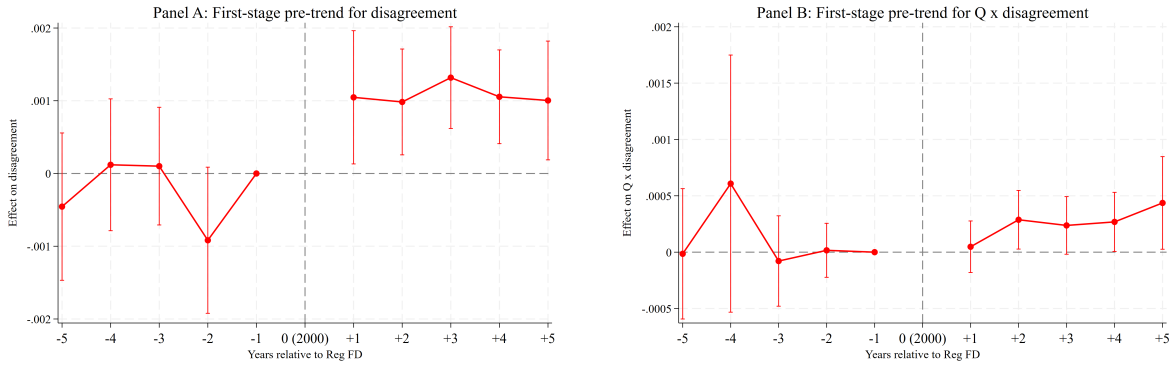


Figure 3: Parallel Trends for Disagreement and its Interaction with Q around RegFD

This figure plots the coefficients on the variables used to instrument investor disagreement ($DISA_{it}$), and its interaction with Q ($DISA_{it} \times Q_{it-1}$). The left panel plots the coefficients on the interactions between $Treat$ and event-time indicators for years relative to Reg FD, using $t = -1$ as the omitted reference period. These coefficients capture the dynamic effect of Reg FD on $DISA_{it}$. The right panel plots the coefficients on the interactions between $Treat$, Q_{it-1}, and the same event-time indicators. These coefficients capture changes in $DISA_{it} \times Q_{it-1}$ around Reg FD.

Appendix A: Variable Definition

Table A1: Variable Definitions

Variable	Definition
Investment, disagreement, and control variables	
INV	A generic investment outcome, measured using $CAPXRNDA$, $CAPXA$, or $RNDA$, depending on the specification.
$CAPXRNDA$	The sum of capital expenditures and research and development expenses, scaled by lagged total assets.
$CAPXA$	Capital expenditures scaled by lagged total assets.
$RNDA$	Research and development expenses scaled by lagged total assets.
Q	Tobin's Q, calculated as the market value of equity plus the book value of total assets minus the book value of equity, scaled by total assets.
$DISA$	Investor disagreement, calculated as the natural logarithm of one plus the cross-sectional standard deviation of analysts' EPS forecasts scaled by the stock price.
$ANALYST$	The natural logarithm of the number of analysts covering the firm.
$SIZE$	Firm size, measured as the natural logarithm of total assets.
CF	Cash flow, measured as the sum of income before extraordinary items, depreciation and amortization expenses, and research and development expenses, scaled by lagged total assets.
$LEVERAGE$	The sum of debt in current liabilities and long-term debt, scaled by total assets.
IO	Institutional ownership, measured as the number of shares held by 13F institutional investors divided by total shares outstanding.
RET	The annualized value-weighted market-adjusted cumulative stock return over the subsequent three years, beginning at the end of the investment year.
$INVASSET$	The inverse of lagged total assets.
Earnings-announcement and Regulation FD variables	
$DISA^{rev}$	Revision-based disagreement. For each interim earnings announcement, it is calculated as the cross-sectional standard deviation of analysts' annual EPS forecast revisions, where a revision equals the matched post-announcement forecast minus the pre-announcement forecast, scaled by the stock price. Pre-announcement forecasts are issued within 45 days before the announcement, and post-announcement forecasts are issued by the same analysts within 30 days after the announcement. The annual measure is the average across the three interim fiscal-quarter announcements.
$DISA^{pre}$	Pre-announcement forecast dispersion, measured as the cross-sectional standard deviation of matched analysts' annual EPS forecasts issued within 45 days before an interim earnings announcement, scaled by the stock price and averaged across the interim announcements within the fiscal year.
$DISA^{post}$	Post-announcement forecast dispersion, measured as the cross-sectional standard deviation of matched analysts' annual EPS forecasts issued within 30 days after an interim earnings announcement, scaled by the stock price and averaged across the interim announcements within the fiscal year.
$ANALYST^{rev}$	The natural logarithm of the number of analysts who issue both pre- and post-announcement EPS forecasts used to construct $DISA^{rev}$.
$Treat$	An indicator equal to one for firms whose analyst forecast accuracy in fiscal year 1999 is above the sample median, and zero otherwise. Forecast accuracy is defined as the negative absolute difference between the consensus analyst EPS forecast and actual EPS, scaled by the stock price.
$Post$	An indicator equal to one for fiscal years 2001 through 2005 and zero for fiscal years 1995 through 1999. Fiscal year 2000 is excluded from the main difference-in-differences estimation.
Cross-sectional and mechanism variables	

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Table A1 continued from the previous page

Variable	Definition
BM	The book-to-market ratio, measured as the book value of equity divided by the market value of equity.
FC	Financial constraints, measured as capital expenditures minus cash flow, scaled by capital expenditures. Higher values indicate greater reliance on external financing and, under the classification used in the paper, lower financial constraints.
$Cov(V, T)$	The firm-year covariance between squared daily stock returns and daily share turnover. Firms with above-median values within a fiscal year are classified as rational-expectations firms, for which investors are more likely to condition on prices, while firms with below-median values are classified as differences-of-opinion firms.
$EarnVol$	Earnings volatility, measured as the standard deviation of income before extraordinary items available to common shareholders scaled by total assets over the preceding five fiscal years. At least three annual observations are required.
$Complexity$	The firm-complexity measure developed by Loughran and McDonald (2024), constructed from complexity-related language in firms' 10-K filings.
$\widehat{DISA}$	The valuation-uncertainty-driven component of investor disagreement, obtained as the predicted value from a panel regression of $DISA$ on either $EarnVol$ or $Complexity$, including firm and year fixed effects.
$DISA^{\perp}$	The component of investor disagreement orthogonal to valuation uncertainty, measured as the residual from the panel regression used to obtain $\widehat{DISA}$.
$PeerQ$	The aggregate Tobin's Q of peer firms in the same industry. Industries are defined using either the FIC-100 text-based industry classification of Hoberg and Phillips (2016) or two-digit SIC codes. Peer Q is calculated using the equally weighted mean, the firm-size-weighted mean, or the industry median.
Alternative disagreement measures	
$DISA_BSD$	Institutional buyer–seller disagreement, calculated as $-\ln\left(\left\lvert\frac{Buyers-Sellers}{Buyers+Sellers}\right\rvert + 0.0001\right)$, where $Buyers$ and $Sellers$ are the numbers of 13F institutions that increase and decrease, respectively, their holdings of the stock during the quarter.
$DISA_TD$	Institutional trading dispersion, calculated as $\ln\left(\frac{\mathrm{SD}_j(\Delta Holdings_{j,i,t})}{Shrout_{i,t}}\right)$, where $\Delta Holdings_{j,i,t}$ is the quarterly change in the number of shares of stock i held by institution j, and $Shrout_{i,t}$ is the firm's split-adjusted shares outstanding.
$DISA_Mean$	Analyst forecast dispersion measured as the cross-sectional standard deviation of analysts' EPS forecasts scaled by the mean analyst EPS forecast.
$DISA_ATS$	Analyst forecast dispersion measured as the cross-sectional standard deviation of analysts' EPS forecasts scaled by assets per share.
$DISA_SUV$	Standardized unexplained trading volume, following Huang et al. (2021). Monthly log turnover is first filtered using a rolling AR(4) model. The unexplained turnover component is then regressed on the positive and negative components of stock returns over a rolling 60-month window, requiring at least 36 observations. The monthly measure is the standardized regression residual, and the annual measure is its average over the 12 months ending in the firm's fiscal year-end month, requiring at least six valid monthly observations.
Private information measures	
$Nonsync$	Price nonsynchronicity, calculated as $\ln\left(\frac{1-R^2}{R^2}\right)$, where R^2 is obtained from a stock-level time-series regression of monthly excess returns on the Fama and French (1993) three factors. The measure is estimated using the preceding 60 months and requires at least 36 monthly observations.
PIN	The probability of informed trading developed by Easley et al. (1996), which uses buyer- and seller-initiated order flow to estimate the probability that trading originates from privately informed investors.
$GPIN$	Generalized probability of informed trading, which extends PIN by allowing greater flexibility in uninformed trading intensity and is designed to distinguish private-information arrival more effectively from noise-driven trading activity.

Continued on the next page

Table A1 continued from the previous page

Variable	Definition
PrivateInfo	The private-information share developed by Brogaard et al. (2022), measured as the proportion of stock-return variance attributable to private firm-specific information revealed through trading. It is estimated separately for each firm-year using daily observations and the permanent price impact of shocks to signed trading volume.